47-851: Modern Convex Optimization Lecture 0

Modern Convex Optimization

Lecturer: Karan Singh

Contents

1. A Motivating Example 3
2. Preliminaries and Conventions 4
   2.1. When do minimizers exist? 5
3. Convexity of Sets and Functions 6
4. Local and Global Minimizers 8
5. Computational Hardness of Local Minimization 9
   5.1. Global problems can be difficult 9
   5.2. Embedding global hardness locally 10
6. First-Order Conditions for Local Minimization 12
   6.1. The Constrained Case 13
7. Positive Semidefinite Matrices 14
8. Second-Order Conditions for Local Minimization 16
   8.1. Why not the constrained case? 16
9. A Few Topological Definitions 17
10. The Separation Theorem 18
11. Equivalent Conditions for Convexity 20
   11.1. First-Order Local-to-Global for Convex Functions 21
   11.2. Subgradients and Their Existence 22
12. From Linear Programs to Conic Programs 24
13. A Hierarchy of Conic Programs 26
   13.1. The General Form of Second-Order Conic Programs 27
   13.2. The General Form of Semi-Definite Programs 27
   13.3. Back to The Hierarchy 28
14. Deriving SDP Duals 29
   14.1. The Weak Minimax Theorem 29
   14.2. Weak Duality for SDPs 29
15. Application: Approximation Algorithms for MaxCut 31
   15.1. Cuts and Laplacians 31
   15.2. Nesterov's Approximation for BQPs 31
16. Application: Stabilizability of Linear Dynamical Systems 33
   16.1. Lyapunov Stability 33
   16.2. Moving onto Stabilizability 35
17. Conic Duality 36
   17.1. Dual Cones and Weak Conic Duality 36

17.2. Proving Strong Conic Duality 38
18. The S-Lemma 39
18.1. The Homogenous Case 40
18.2. Finishing up the General Case 41
Q1. Inner Products and Dual Norms (10 points) 44
Q2. Local Descent and Checking PSD (10 points) 44
Q3. Quasiconvexity and Back (10 points) 44
Q4. Eigenvalues and SDPs (15 points) 46
Q5. Lovász's bound on Shannon capacity (15 points) 46
Q6. Grads and Dists (10 points) 48
Q7. Just Enough Convexity (10 points) 48
Q8. Small Grad Bounds (10 points) 48

47-851: Modern Convex Optimization Lecture 1

Introduction and Complexity of Local Minimization

Lecturer: Karan Singh

This is a fast-paced course that develops the basic toolkit of convex optimization. We will see what makes convexity an attractive lens to examine optimization problems, and develop efficient algorithms. A good part of the course will also be spent on (conceptual) applications in applied mathematics and engineering. In roughly five weeks, we will visit:

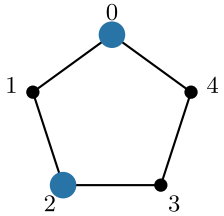
1. Convex sets and functions
2. Duality: Conic, Lagrangian, Fenchel
3. Structured conic optimization: theory, models, applications
4. Large-scale optimization: ellipsoid, first-order methods, acceleration, lower bounds

Contents

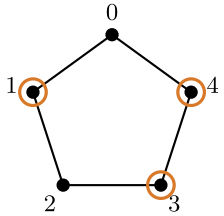
1. A Motivating Example . . . 3
2. Preliminaries and Conventions . . . 4
   2.1. When do minimizers exist? . . . 5
3. Convexity of Sets and Functions . . . 6
4. Local and Global Minimizers . . . 8
5. Computational Hardness of Local Minimization . . . 9
   5.1. Global problems can be difficult . . . 9
   5.2. Embedding global hardness locally . . . 10

1. A Motivating Example

Consider a language or a script with n alphabets. Some letter pairs look alike. If one were to communicate using alphabets alone, one must avoid the use of at least one alphabet in all such letter pairs. How many letters are usable?



A stable set: $\{0, 2\}$



Its complement is a vertex cover

Consider a graph $G = (V, E)$ where V is the set of letters and $(u, v) \in E$ if and only if u and v look alike; this notion is symmetric. We are looking for the largest *stable set*.

$$\alpha(G) = \max_{S \subseteq V} \{|S| : S \text{ contains at most one endpoint of every edge}\}.$$

Other related graph-theoretic notions:

- Clique is a subset of vertices that are all pairwise connected.
- Vertex Cover is a subset of vertices such that for every edge, at least one endpoint is in it.

Exercise. Prove that S is an independent set on G iff S is a clique on the edge-complement of G.

Exercise. Prove that S is an independent set on G iff $V - S$ is a vertex cover on G.

Another interesting fact is that the maximum number of maximum (in fact, maximal too) independent sets is $3^{\frac{n}{3}} \ll 2^n$. Can you think of graphs for which this is attained?

A question of Claude Shannon (1956) concerns "worst-case information theory": How much more efficient are words than letters? If one were to communicate with k-letter words, $|V| = n^k$, and 2 words $(\alpha_1 ... \alpha_k)$, $(\beta_1, ... \beta_k)$ would be ambiguous if and only if they are not the same, yet for all positions $1 \leq \ell \leq k$, α_ℓ and β_ℓ are ambiguous (or the same). Drawing edges between such words, we get G^k. Graph theorists call it the "strong graph product," denoted by $\boxtimes$, applied k times.

Exercise. How do the adjacency matrices of G and G^k relate? Think of Kronecker products.

We expect $\alpha\big(G^k\big)$ to scale exponentially in k; after all the vertex set does. So to normalize this effect, let us define

$$\Theta(G) = \lim_{k \to \infty} \sqrt[k]{\alpha(G^k)}.$$

Exercise. Prove that $\alpha\big(G^k\big) \geq (\alpha(G))^k$.

Hint: Using unambiguous letters alone one can construct unambiguous words.

But in fact, $\Theta(G)$ can be quite a bit greater than $\alpha(G)$, indicating words are more communication-efficient than letters. Here was the concrete question Shannon posed.

C_n is the cycle graph with n vertices. Clearly $\alpha(C_5) = 2$. How large is $\Theta(C_5)$?

This was resolved by Laszlo Lovasz, in 1979, 23 years later in a result that is one of many origins of Semi-Definite Programming; this we will study, and you'll prove the result in one of the assignments!

Theorem 1 (Lovasz '79) Using SDPs, we have

$$\Theta(C_5) = \sqrt{5} > 2.$$

An unresolved question even in the LLM age is: what is $\Theta(C_7)$?

2. Preliminaries and Conventions

The following are notions of *similarity* and *length* respectively.

Definition 2 $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$, where V is a vector space, is an inner product if
1. $\langle x, x \rangle \geq 0$ for all $x \in V$ and $\langle x, x \rangle = 0$ if and only if $x = 0$.
2. $\langle x, y \rangle = \langle y, x \rangle$ for all $x, y \in V$.
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ for all $x, y \in V$ and $\alpha \in \mathbb{R}$.
4. $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ for all $x, y, z \in V$.

Definition 3 $\|\cdot\| : V \to \mathbb{R}$, for vector space V, is a norm if
1. $\|x\| \geq 0$ for all $x \in V$ and $\|x\| = 0$ if and only if $x = 0$.
2. $\|\alpha x\| = |\alpha| \, \|x\|$ for all $x \in V$ and $\alpha \in \mathbb{R}$.
3. $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$.

If the second part of condition (1) fails in either definition, we call these semi inner products and semi norms respectively. Under such definitions, non-zero vectors can have zero *length*.

Lemma 4 (Cauchy-Schwarz) For any (semi) inner product, for all $x, y \in V$

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle.$$

Proof. Consider that $\|sx - ty\|^2 \geq 0$ for all $x, y \in V$. We will fix s, t later.

$$s^2 \langle x, x \rangle + t^2 \langle y, y \rangle - 2st \langle x, y \rangle \geq 0$$

$$\Rightarrow \frac{s^2 \langle x, x \rangle + t^2 \langle y, y \rangle}{2} \geq st \langle x, y \rangle.$$

Substitute $s = \frac{1}{\sqrt{\langle x, x \rangle}}$, $t = \frac{1}{\sqrt{\langle y, y \rangle}}$, then we are done unless $\langle x, x \rangle$ or $\langle y, y \rangle$ is zero. □

Exercise. Complete the above proof for cases when $\langle x, x \rangle$ or $\langle y, y \rangle$ is zero.

Hint: Think about what happens if we choose arbitrarily large s or t in these cases.

Proposition 5 $f(x) = \sqrt{\langle x, x \rangle}$ is a norm.

Proof. (1) & (2) are straightforward. For all $x, y \in V$, we have

$$\begin{aligned} f(x + y)^2 &= \langle x, x \rangle + \langle y, y \rangle + 2 \langle x, y \rangle \\ &\leq \langle x, x \rangle + \langle y, y \rangle + 2 f(x) f(y) \quad \text{(C.S.)} \\ &\leq (f(x) + f(y))^2. \end{aligned}$$

□

2.1. When do minimizers exist?

For practical optimization problems, a desirable property is that there is indeed a best decision, no matter how difficult or easy it is to find.

Definition 6 A set S is closed if for any (convergent) sequence contained within it, the corresponding limit also lies in it.

A set S is bounded if $\exists C \in \mathbb{R}$ such that $\forall x, y \in S$, $\|x - y\| \leq C$.

Compact = bounded and closed.

A generic optimization problem looks like

$$\mathcal{O}(f, K) \equiv \min_{x \in K} f(x).$$

Definition 7 $\mathcal{O}(f, K)$ is feasible if K is not empty.

$\mathcal{O}(f, K)$ is bounded below if $\exists C \in \mathbb{R}$, such that $\forall x \in K, f(x) \geq C$.

$\mathcal{O}(f, K)$ is solvable if $\exists x^* \in K$ such that $\forall x \in K, f(x) \geq f(x^*)$.

For linear programming:

$$\text{feasible \& bounded below} \Rightarrow \text{solvable.}$$

But **false** for Convex programs: $\min_{x \in \mathbb{R}} e^{-x}$.

This is the cause of a lot of pain when stating any non-LP duality result.

Theorem 8 If K is compact & non-empty and f is continuous, then a minimizer exists.

This theorem is nice, but we would also like something that applies to unconstrained problems.

Theorem 9 If K is closed & non-empty, f is continuous with all its sub-level sets bounded, then a minimizer exists.

Definition 10 α-sublevel set of f on K for any $\alpha \in \mathbb{R}$ is $S_\alpha(f) = \{x \in K : f(x) \leq \alpha\}$.

3. Convexity of Sets and Functions

Consider V to be any vector space and K to be any convex set so that adding elements makes sense.

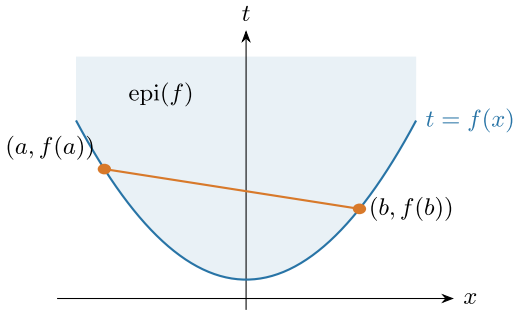
Definition 11 (Convex Set) $S \subseteq V$ is convex if $\forall x, y \in S$, $\lambda \in [0, 1]$, $\lambda x + (1 - \lambda) y \in S$.

Definition 12 (Convex Function) $f : K \to \mathbb{R}$ is convex if $\forall x, y \in K$, $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda) y) \leq \lambda f(x) + (1 - \lambda) f(y).$$

The set $\{\lambda x + (1-\lambda)y : \lambda \in [0,1]\}$ is the line segment connecting x and y. So convex sets are precisely those that contain all line segments for which corresponding endpoints are within the set. And convex functions are precisely those for which any linear interpolation overapproximates the function value. How are these related?

Definition 13 (Epigraph) $\text{epi}(f) = \{(t,x) : t \geq f(x)\}$.



Theorem 14 f is a convex function if and only if $\text{epi}(f)$ is a convex set.

Proof. $\Rightarrow$ Take $(t,x),(s,y) \in \text{epi}(f)$. Then

$$\lambda(t,x) + (1-\lambda)(s,y) = (\lambda t + (1-\lambda)s, \lambda x + (1-\lambda)y) \in \text{epi}(f)$$

since

$$\begin{aligned} \lambda t + (1-\lambda)s &\geq \lambda f(x) + (1-\lambda)f(y) \quad \text{by membership} \\ &\geq f(\lambda x + (1-\lambda)y). \end{aligned}$$

$\Leftarrow$ Take $x, y \in K$, $\lambda \in [0,1]$. Note that $(f(x),x),(f(y),y) \in \text{epi}(f)$. Hence

$$(\lambda f(x) + (1-\lambda)f(y), \lambda x + (1-\lambda)y) \in \text{epi}(f).$$

Thus $\lambda f(x) + (1-\lambda)f(y) \geq f(\lambda x + (1-\lambda)y)$. □

One attraction of convexity is that it is a robust notion, that is, it is preserved under many (useful) operations. We will describe a few operations for convex sets below. You should be able to translate these to convex functions via the above link.

Exercise. Prove that Cartesian products, Minkowski sums, linear transformations and intersections of convex sets are convex.

If you have not seen this before, $A + B = \{a + b : a \in A, b \in B\}$ is a Minkowski sum.

Exercise. Construct a closed set $S \subseteq \mathbb{R}^n$, and a matrix $A \in \mathbb{R}^{m \times n}$ such that the linear transformation $AS = \{Ax : x \in S\}$ is not closed.

Hint: Linear transformations can be used to do projections onto subspaces.

Exercise. Prove $S_\alpha(f)$ is a convex set $\forall \alpha \in \mathbb{R}$ if f is a convex function. Is the converse true?

4. Local and Global Minimizers

Definition 15 $\overline{x}$ is a local minimum for f on K if $\exists \varepsilon > 0$ such that

$$f(\overline{x}) \leq f(x) \quad \forall x \in K \cap B(\overline{x}, \varepsilon).$$

$\overline{x}$ is a strict local minimum for f on K if $\exists \varepsilon > 0$ such that

$$f(\overline{x}) < f(x) \quad \forall x \in K \cap B(\overline{x}, \varepsilon) - \{\overline{x}\}.$$

$\overline{x}$ is a global minimum for f on K if $\forall x \in K$

$$f(\overline{x}) \leq f(x).$$

$\overline{x}$ is a strict global minimum for f in K if

$$\forall x \in K - \{\overline{x}\}, \quad f(\overline{x}) < f(x).$$

Here and elsewhere, $B(\overline{x}, \varepsilon) = \{x : \|x - \overline{x}\| \leq \varepsilon\}$. Think of $\|\cdot\|$ as the Euclidean norm, but for finite-dimensional vector spaces, all norms are equivalent (sandwich each other up to constants), so it does not matter. These definitions apply to any (for example, non-convex) function. Also note that global minima are local minima.

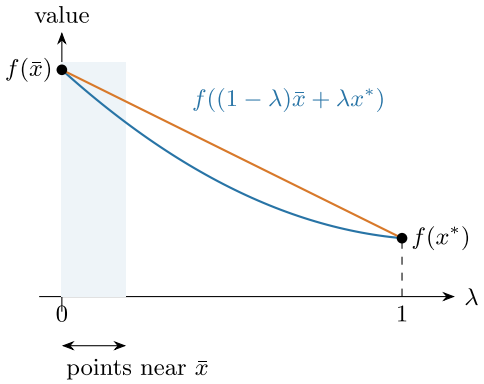
Exercise. Give a non-constant function over $K = \mathbb{R}$ all of whose local minima are global *maxima*.

Proposition 16 For a convex function f, any local minimum is also a global minimum.

Proof. Take a local minimum $\overline{x} \in K$. If $\overline{x}$ is not a global minimum, then $\exists x^* \in K$ such that $f(x^*) < f(\overline{x})$. Note x^* need not be a global minimum; a priori it is not clear if the latter exists. Now set $x = (1 - \lambda)\overline{x} + \lambda x^* = \overline{x} + \lambda(x^* - \overline{x}) \in K$; we will ensure $\lambda \in (0, 1]$. We have

$$f(x) \leq (1 - \lambda) f(\overline{x}) + \lambda f(x^*) < f(\overline{x}),$$

yet x can be made arbitrarily close to $\overline{x}$. For example, if we want the distance to be at most ε, we can set $\lambda = \min\left\{1, \frac{\varepsilon}{\|\overline{x} - x^*\|}\right\}$. □



You might expect that finding a local minimum is somewhat easy: given that it imposes only local qualifications, maybe some greedy procedure produces these. If so, for convex functions, the above

proposition translates them to global minima. This is a key attraction of convexity: doing well enough locally ensures global performance. But this intuition is misleading, as we show next.

5. Computational Hardness of Local Minimization

Theorem 17 Given $\overline{x} \in \mathbb{R}^d$, checking if $\overline{x}$ is a local minimum for a degree 4 polynomial is NP-hard.

Theorem 18 Given $\overline{x} \in \Delta$, where $\Delta = \left\{x \in \mathbb{R}^n : x \geq 0, \sum_{i=1}^n x_i \leq 1\right\}$, checking if $\overline{x} \in \Delta$ is a local (constrained) minimum for a degree 2 polynomial is NP-hard.

We will prove a weaker result.

Theorem 19 (STRICT-4-HARD) Checking if $x \in \mathbb{R}^n$ is a local strict minimum for a degree 4 polynomial is NP-hard.

Note that these results concern certification of a given point. It can still be that other local minima are easy to find; this does happen for a few problem classes. But also finding something that is guaranteed to exist (via compactness, say) cannot be NP-hard, which concerns decision problems, not search. These problems are better translated to PPAD-hardness.

5.1. Global problems can be difficult

We will build towards a proof of the last theorem. The easy bit is to believe that global minimization is hard; so let us start there.

Definition 20 **3SAT** is NP-complete and asks if $\exists$ a boolean assignment of $x, y, z...$ such that

$$(x \vee \overline{y} \vee z) \wedge (\overline{x} \vee y \vee \overline{z}) \wedge ...$$

is satisfied. Three literals in each clause are joined by ORs with negations allowed. Clauses themselves are joined by ANDs.

1-in-3SAT is also NP-complete. It asks if there is a boolean assignment that makes exactly one of the literals in each OR clause true.

***d*-POLYPOS** asks if given a degree-d multivariate polynomial (d is a fixed number, so no representational issues here), is it true that $\forall x \in \mathbb{R}^n, P(x) > 0$.

Claim 21 6-POLYPOS is NP-hard.

Proof. We will embed 3SAT in a degree 6-poly.

$$P(x) = \sum_{i=1}^{n} (x_i(1-x_i))^2$$
$$+\underbrace{((x_1 + (1-x_2) + x_3 - 1)(x_1 + (1-x_2) + x_3 - 2)(x_1 + (1-x_2) + x_3 - 3))^2}_{x \vee \overline{y} \vee z} + \ldots$$

There is a degree 6 term for each OR clause.

$P(x) = 0$ if and only if $x_i \in \{0, 1\}$ and for each clause, at least one (signed) literal is true, e.g.,

$$x_1 + (1 - x_2) + x_3 \in \{1, 2, 3\}$$

which is the same as 3SAT being satisfied. □

Claim 22 4-POLYPOS is NP-hard.

Proof. Same proof, except we embed 1-in-3SAT. This time we only need $(x_1 + (1 - x_2) + x_3 - 1)^2$. □

We are butchering complexity-theoretic reductions a bit. We are using Cook/algorithmic reductions, whereas Karp/logical reductions are more standard. But Cook-NP-hard is plenty hard too.

5.2. Embedding global hardness locally

For any degree-d polynomial P, let $P_h(x, y) := y^d P\left(\frac{x}{y}\right)$ be its homogeneous counterpart. Note that $P_h(x, 1) = P(x)$. Also note that this transformation is symbolic; so we can plug in $y = 0$.

Observation: For any homogeneous polynomial P_h, $\forall \alpha \geq 0$, $P_h(\alpha x) = \alpha^d P_h(x)$, where d is the degree.

Claim 23 For a homogeneous poly q, $x = 0$ is a strict local min $\Leftrightarrow q(x) > 0$ for all $x \neq 0$.

Proof. $\Leftarrow$ $q(0) = 0$ always. We are done.

$\Rightarrow$ $\exists \varepsilon > 0$ such $\forall y \in B(0, \varepsilon) - \{0\}$, $q(y) > q(0) = 0$. But then consider any $z \in \mathbb{R}^n$, $z \neq 0$.

$$q(z) = q\left(\frac{\|z\|}{\varepsilon} \times \frac{z}{\|z\|}\varepsilon\right) = \left(\frac{\|z\|}{\varepsilon}\right)^d q\left(\frac{z}{\|z\|}\varepsilon\right) > 0$$

because $\frac{z}{\|z\|}\varepsilon$ belongs to $B(0, \varepsilon)$. □

The reason for considering homogenized polynomials is as follows: they display the same quantitative behavior in a neighborhood of the origin as they display elsewhere, up to suitable scaling. This characteristic can be used to hide global hardness in a small near-zero neighborhood; the latter is what local minima care about.

Finally, we know that testing positivity of degree 4 polynomials is NP-hard. If this were also established for homogeneous equivalents, we would be done. Except this doesn't make sense directly because homogeneous polynomials evaluate to zero at zero. To complete this translation, consider the following assertion for even degree polynomials:

$$P(x) > 0 \quad \forall x \quad \Leftrightarrow \quad P_h(x, y) > 0 \quad \forall (x, y) \neq 0.$$

To be clear, we will fail to prove this as stated, but we will establish this for special cases which will nevertheless suffice for us.

Here, $\Leftarrow$ holds always as $P(x) = P_h(x, 1) > 0$.

$\Rightarrow$ is tricky. Let's try the contrapositive.

$$\exists (x, y) \neq 0 \text{ such } P_h(x, y) \leq 0.$$

We wish to conclude $\exists z$, $P(z) \leq 0$. This is indeed fine if $y \neq 0$ incidentally.

$$P\left(z := \frac{x}{y}\right) = \frac{P_h(x, y)}{y^d} \leq 0.$$

But this doesn't work if $y = 0$. In fact, consider the counterexample below.

Example. $P(x_1, x_2) > 0$ everywhere, yet $P_h(1, 0, 0) = 0$, where

$$P(x_1, x_2) = x_1^2 + (1 - x_1 x_2)^2,$$

$$P_h(x_1, x_2, y) = y^2 x_1^2 + \left(y^2 - x_1 x_2\right)^2.$$

The subtle observation is that the bad case never happens for the polynomial constructed for 1-in-3SAT. Indeed

$$P_h(x, y) = \sum_{i=1}^{n} \left(x_i(y - x_i)\right)^2 + \left(y(x_1 + (y - x_2) + x_3) - y^2\right)^2 + ...$$

With $y = 0$, the first term becomes $\sum_{i=1}^{n} x_i^4$. Hence $P_h(x, 0) \leq 0 \Rightarrow x_i = 0$ for all i. But the only problematic case was when $\exists (x, y) \neq 0$ and $y = 0$ where $P_h(x, y) \leq 0$.

So our proof of the hardness of certifying strict local minima is complete.

47-851: Modern Convex Optimization Lecture 2

Convexity and Separation

Lecturer: Karan Singh

So far, we have seen that an attractive property of convex functions is that local minima are also global minima. However, we have also seen that even local minima can be hard to find and certify generally. So we will begin by relaxing the definition of local minima itself, in a way that retains the local-to-global property while being tractable.

Contents

6. First-Order Conditions for Local Minimization 12
6.1. The Constrained Case 13
7. Positive Semidefinite Matrices 14
8. Second-Order Conditions for Local Minimization 16
8.1. Why not the constrained case? 16
9. A Few Topological Definitions 17
10. The Separation Theorem 18
11. Equivalent Conditions for Convexity 20
11.1. First-Order Local-to-Global for Convex Functions 21
11.2. Subgradients and Their Existence 22

6. First-Order Conditions for Local Minimization

Definition 24 d is a descent direction of f at x if there exists $\overline{\alpha} > 0$ such that $f(x + \alpha d) < f(x)$ for all $\alpha \in (0, \overline{\alpha})$.

Claim 25 If $\langle \nabla f(x), d \rangle < 0$, then d is a descent direction for f at x.

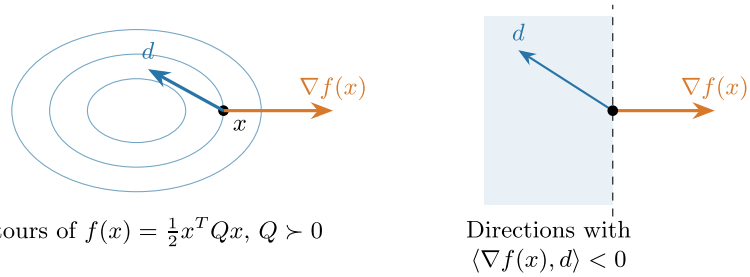
Proof. By Taylor's theorem, we have

$$f(x + \varepsilon d) = f(x) + \langle \nabla f(x), d \rangle \varepsilon + r(\varepsilon, x, d),$$

where $r(x, \varepsilon, d) = o(\varepsilon)$, that is, $\lim_{\varepsilon \to 0^+} r(\varepsilon, x, d)/\varepsilon = 0$. There exists $\overline{\varepsilon} > 0$ such that for all $0 < \varepsilon \leq \overline{\varepsilon}$, $|r(\varepsilon, x, d)/\varepsilon| \leq |\langle \nabla f(x), d \rangle| \, /2$. Then, we get

$$\frac{f(x + \varepsilon d) - f(x)}{\varepsilon} = \langle \nabla f(x), d \rangle + \frac{r(\varepsilon, x, d)}{\varepsilon} \leq \frac{1}{2} \langle \nabla f(x), d \rangle < 0.$$

□



Contours of $f(x) = \frac{1}{2}x^T Qx$, $Q \succ 0$ | Directions with $\langle \nabla f(x), d \rangle < 0$

The converse is false. Consider $f(x) = -x^2$; then at 0 any non-zero direction is a descent direction.

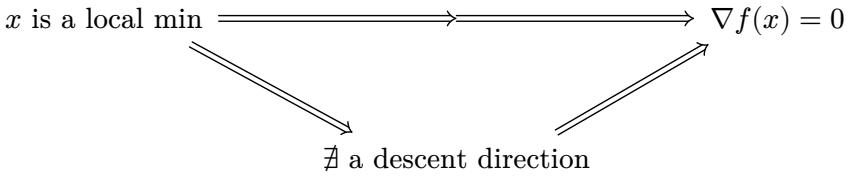
Proposition 26 *x is a local min for $f \Rightarrow \nabla f(x) = 0$.*

Proof. Say $\nabla f(x) \neq 0$; then $\langle \nabla f(x), -\nabla f(x) \rangle < 0$. Hence $-\nabla f(x)/\|\nabla f(x)\|$ is a descent direction. Hence there exists $\overline{\alpha} > 0$ such that for all $0 < \varepsilon < \overline{\alpha}$, there exists $x' \in B(x, \varepsilon)$, namely

$$x' = x - \varepsilon \frac{\nabla f(x)}{\|\nabla f(x)\|},$$

such that $f(x') < f(x)$. Hence x is not a local min. □

To summarize all this, we have:



Exercise. Provide separate counterexamples for each of the three converses.

6.1. The Constrained Case

Now, let's do the constrained case. We have dealt with non-convex functions so far. Coming to the constrained case, you might be tempted to also treat general (non-convex) sets. But, for example, if K is the set of vertices on a hypercube, all feasible points are local minima, regardless of the objective function. To avoid such pathologies, it makes sense to only consider a convex set K.

Proposition 27 *$\overline{x}$ is a local min for f on a convex set $K \Rightarrow \langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0$ for all $x \in K$.*

Proof. Say there exists $x \in K$, $\langle \nabla f(\overline{x}), x - \overline{x} \rangle < 0$. Then consider

$$y = (1 - \lambda)\overline{x} + \lambda x = \overline{x} + \lambda(x - \overline{x}) \in K.$$

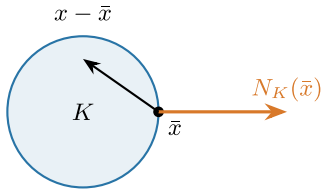
Now, by Taylor's theorem, $f(y) = f(\overline{x}) + \lambda \langle \nabla f(\overline{x}), x - \overline{x} \rangle + r(\lambda, x, \overline{x})$ where $r(\lambda, x, \overline{x}) = o(\lambda)$. Then there exists $\overline{\lambda} > 0$ such that for all $0 < \lambda < \overline{\lambda}$, $|\frac{r(\lambda, x, \overline{x})}{\lambda}| < \frac{1}{2} |\langle \nabla f(\overline{x}), x - \overline{x} \rangle|$. For all such λ's, we have

$$f(y) \leq f(\overline{x}) + \frac{\lambda}{2} \langle \nabla f(\overline{x}), x - \overline{x} \rangle < f(x).$$

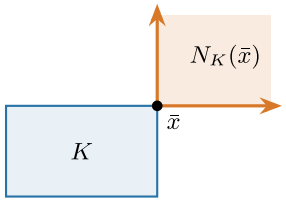
Thus, there does not exist an $\varepsilon > 0$ such that $f(y) \geq f(\overline{x}) \forall y \in B(\overline{x}, \varepsilon)$, because $\lambda(x - \overline{x})$ can be made arbitrarily small, yet $f(\overline{x} + \lambda(x - \overline{x})) < f(\overline{x})$. □

The normal cone of K at x captures all the directions starting at x along which one cannot progress while being restricted to K in a very strong sense, that is, it is not even possible to move along directions which correlate with a member of the cone.

Definition 28 $N_K(\overline{x}) = \{d : \langle d, x - \overline{x} \rangle \leq 0, \forall x \in K\}$ is the normal cone of K at $\overline{x}$.



Smooth boundary: a ray



Corner: a cone of normals

With this definition, we get an alternative characterization:

$$\langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0, \forall x \in K \equiv -\nabla f(\overline{x}) \in N_K(\overline{x}).$$

In words, one would like to go in the direction of $-\nabla f(\overline{x})$ to decrease function value, but all directions that cast a positive shadow on $-\nabla f(\overline{x})$ are unapproachable at $\overline{x}$ by moving in feasible directions.

7. Positive Semidefinite Matrices

Now we have relaxed analogs of local minima which can be easily checked and found. Nevertheless, we have lost something in this relaxation. Can we at least come up with more stringent notions that help us certify local minima in special cases?

To do this, we will consider high-order derivatives, for which the definition below captures the central property of interest.

Definition 29 A symmetric matrix $S \in \mathbb{R}^{n \times n}$ is positive semi-definite if $x^\top S x \geq 0, \forall x \in \mathbb{R}^n$. In this case, we will write $S \succeq 0$. Also $S_+^{n \times n}$ is the space of such matrices.

Similarly, we'll say S is positive definite or $S \succ 0$ if $\forall x \neq 0, x^\top S x > 0$.

Theorem 30 (Spectral Theorem) Any symmetric matrix can be written as

$$S = \sum_{i=1}^{n} \lambda_i u_i u_i^\top$$

where $\lambda_i \in \mathbb{R}$ are eigenvalues, and u_i's are the corresponding eigenvectors (all orthonormal).

Theorem 31 The following statements are all equivalent:
1. $S \succeq 0$.
2. All eigenvalues of S are non-negative.
3. $S = LL^\top$ for some $L \in \mathbb{R}^{n\times k}$ for some k.
4. There exist vectors $v_1, ..., v_n \in \mathbb{R}^k$ for some k such that $S_{ij} = v_i^\top v_j, \forall i, j \in [n]$.
5. Determinants of all principal submatrices are non-negative.

Recall that principal submatrices are those formed by only keeping (or equivalently, deleting) columns and rows corresponding to the same set of indices.

Proof. We will leave out (5), and form a chain from (1) through (4).

(1) $\Rightarrow$ (2) For all x, $x^\top Sx \geq 0$. Take $x = u_i$. Then $0 \leq u_i^\top S u_i = \lambda_i \|u_i\|^2$.

For the next two steps, we reinterpret matrix multiplication two ways. Consider $A \in \mathbb{R}^{n\times k}$, $B \in \mathbb{R}^{k\times n}$; then the standard definition implies

$$(AB)_{ij} = \big(i^{\text{th}} \text{ row of } A\big)^\top \big(j^{\text{th}} \text{ column of } B\big).$$

But matrix multiplication can also be written via outer products as

$$AB = \sum_{i=1}^{k} \big(i^{\text{th}} \text{ column of } A\big)\big(i^{\text{th}} \text{ row of } B\big)^\top.$$

(2) $\Rightarrow$ (3) Hence, $LL^\top = \sum_{i=1}^{n} \big(i^{\text{th}} \text{ column of } L\big)\big(i^{\text{th}} \text{ column of } L\big)^T$. So to mirror the spectral decomposition, we can choose

$$L = \Big[\sqrt{\lambda_1}u_1, \sqrt{\lambda_2}u_2, ..., \sqrt{\lambda_n}u_n\Big],$$

which is valid because eigenvalues are non-negative.

(3) $\Rightarrow$ (4) Here we appeal to the standard interpretation of matrix products. In particular, we have

$$S_{ij} = \big(LL^\top\big)_{ij} = \big(i^{\text{th}} \text{ row of } L\big)^\top \big(j^{\text{th}} \text{ row of } L\big),$$

which means we can simply choose v_i to be the ith row of L.

(4) $\Rightarrow$ (1) Note that $x^\top Sx = \sum_{i,j} x_i x_j S_{ij} = \sum_{i,j} x_i x_j v_i^\top v_j = \left\|\sum_{i=1}^n x_i v_i\right\|^2 \geq 0$. □

Let's try to make sense of (5) by proving its equivalence for 2x2 matrices. Take

$$S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

(5) says $a, c \geq 0$, $ac - b^2 \geq 0$. Recall that λ's are roots of

$$0 = |S - \lambda I| = (a-\lambda)(c-\lambda) - b^2 = \lambda^2 - (a+c)\lambda + (ac - b^2).$$

Note $\lambda_1\lambda_2 = ac - b^2 \geq 0$. Also $\lambda_1 + \lambda_2 = a + c \geq 0$. Hence $\lambda_1, \lambda_2 \geq 0 \Rightarrow S \succeq 0$.

(1) says $S \succeq 0 \Rightarrow e_1^\top S e_1 = a \geq 0, e_2^\top S e_2 = c \geq 0$. Also $\lambda_1, \lambda_2 \geq 0 \Rightarrow \lambda_1\lambda_2 = \det(S) = ac - b^2 \geq 0$.

In fact, the last of these works more generally. One can think of isolating a principal submatrix M as choosing a binary matrix P with exactly one non-zero entry per row so that $M = PSP^\top$. Clearly, if $S \succeq 0$ then $M \succeq 0$, and the determinant of M, that is, the product of its eigenvalues, must be non-negative.

The more magical bit is the opposite direction: if the determinant of $PSP^\top$ is non-negative for arbitrary P, that would be enough to get $S \succeq 0$. (Why?) The principal submatrix condition says that checking this for binary P's with one non-zero entry per row is enough.

8. Second-Order Conditions for Local Minimization

We are back to the unconstrained land. Our aim this time is to get both necessary and sufficient conditions for local minima, although not both at the same time.

Proposition 32 x is a local min for $f \Rightarrow \nabla f(x) = 0$ and $\nabla^2 f(x) \succeq 0$.

Proof. We already know $\nabla f(x) = 0$ is necessary. Say $\nabla f(x) = 0$ and $\nabla^2 f(x) \not\succeq 0$. Then there exists u for which $u^T \nabla^2 f(x) u < 0$ and $\|u\| = 1$. Consider $y = x + \alpha u$.

$$f(x + \alpha u) = f(x) + \langle \nabla f(x), u \rangle \alpha + \frac{1}{2}\alpha^2 u^T \nabla^2 f(x) u + r(\alpha, x, u) \quad \text{where} \quad r(\alpha, x, u) = o(\alpha^2)$$

Hence, there exists $\overline{\alpha} > 0$ such that for all $0 < \alpha < \overline{\alpha}$, $|r(\alpha, x, u)| \leq \frac{1}{4} |\alpha^2 u^T \nabla^2 f(x) u|$. Now

$$f(x + \alpha u) \leq f(x) + \frac{1}{4} u^T \nabla^2 f(x) u \alpha^2 < f(x).$$

Yet, $x + \alpha u$ lives in an arbitrarily small neighborhood of x since this holds for all $0 < \alpha < \overline{\alpha}$. Thus, x cannot be a local min. □

Proposition 33 $\nabla f(x) = 0$ and $\nabla^2 f(x) \succ 0 \Rightarrow x$ is a strict local min at x.

Proof. Consider any $y = x + d\varepsilon$ with $\varepsilon \neq 0$ and unit vector d.

$$f(y) = f(x) + \varepsilon \langle \nabla f(x), d \rangle + \frac{\varepsilon^2}{2} d^T \nabla^2 f(x) d + r(x, \varepsilon, d) \quad \text{where} \quad r = o(\varepsilon^2)$$

For any fixed d, there exists $\overline{\varepsilon} > 0$ such that for all $0 < \varepsilon < \overline{\varepsilon}$ we have $|r(x, \varepsilon, d)| \leq \frac{\varepsilon^2}{4} |d^T \nabla^2 f(x) d|$. So $f(y) \geq f(x) + \frac{\varepsilon^2}{4} d^T \nabla^2 f(x) d > f(x)$. The one complication is that this choice of $\overline{\varepsilon}$ depends on d.

To fix this, define $g(d) = \sup\{\overline{\varepsilon} : \forall 0 < \varepsilon < \overline{\varepsilon}, f(x + \varepsilon d) > f(x)\}$, which is continuous. Clearly $g(d) > 0$ for any d; compactness of the set of all unit-norm d's implies that $\min_{\|d\|=1} g(d)$ exists, and hence, is positive. Thus, $f(y) > f(x)$ for all $y \in B\Big(x, \min_{\|d\|=1} g(d)\Big) - \{x\}$. □

8.1. Why not the constrained case?

It is also possible to define second-order conditions (sufficient and necessary) for constrained local minima. However, their practical utility is limited by concerns about computational complexity.

For example, is $x = 0$ a local min for some $f(x)$ on the non-negative orthant $\mathbb{R}_{\geq 0} = \{x : x \geq 0\}$?

To establish necessary conditions, one would hope for:
1. $\langle \nabla f(0), x \rangle \geq 0$ for all $x \in \mathbb{R}_{\geq 0}$ which is the same as $\nabla f(0) \geq 0$.
2. And for all $d \geq 0$ such that $\langle \nabla f(0), d \rangle = 0$, we have $d^\top \nabla^2 f(0) d \geq 0$.

The intuition behind (2) is that for all directions one could move from $x = 0$ without changing the first-order expansion, the second-order term is non-negative.

But, also, checking given a matrix A if for all $x \geq 0$, $x^\top A x \geq 0$ is NP-hard, which is something you would need to do with $\nabla f(0) = 0$. This is called the copositivity problem. Copositive matrices are of course a superset of PSD matrices, but they only attest to non-negativity for non-negative vectors, and checking membership within this set is computationally hard.

9. A Few Topological Definitions

Definition 34 Closure of a set $\mathrm{cl}(A)$ is the smallest closed set containing A.

Here *smallest* is defined in the sense of set inclusion. So two sets may not be comparable, yet the smallest in this sense is well defined.

Definition 35 x is an interior point of A if there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq A$.

Interior of a set $\mathrm{int}(A)$ is the set of all interior points of A.

Definition 36 Boundary of a set $\mathrm{bd}(A) = \mathrm{cl}(A) - \mathrm{int}(A)$.

As an example, consider $\mathbb{Q}$, the set of rational numbers. $\mathrm{cl}(\mathbb{Q}) = \mathbb{R}$, $\mathrm{int}(\mathbb{Q}) = \varnothing$ since any small ball around a rational contains an irrational and $\mathrm{bd}(\mathbb{Q}) = \mathbb{R}$.

As a second example, consider the open ball $A = \{x : \|x\| < 1\}$. Then $\mathrm{cl}(A) = \{x : \|x\| \leq 1\}$ is the closed ball, $\mathrm{int}(A) = A$ and $\mathrm{bd}(A) = \{x : \|x\| = 1\}$ is the sphere.

The annoying bit about these definitions is that the (ambient) space you are in matters. Consider the same set $A = \{x : x_1^2 + x_2^2 < 1, x_3 = 0\}$ in $\mathbb{R}^3$. Now, $\mathrm{cl}(A) = \{x : x_1^2 + x_2^2 \leq 1, x_3 = 0\}$ same as before, but $\mathrm{int}(A) = \varnothing$ since any 3d ball cannot be in the flat set A; $\mathrm{bd}(A) = \mathrm{cl}(A)$. Thus, every point in A is now on the boundary. To avoid such nuisances, we will consider notions that are robust to such changes in description.

Recall that A is an affine space if it can be written as $A = x_0 + V$ for some point x_0 and subspace V.

Definition 37 x is a relative interior point of A if there exists $\varepsilon > 0$, $B(x, \varepsilon) \cap \mathrm{aff}(A) \subseteq A$, where $\mathrm{aff}(A)$ is the smallest affine space containing A.

The relative interior of A, $\mathrm{relint}(A)$, is the set of all relative interior points of A.

Definition 38 The relative boundary of a set A is $\text{rbd}(A) = \text{cl}(A) - \text{relint}(A)$.

Thus for $A = \{x : x_1^2 + x_2^2 < 1\}$, $\text{relint}(A) = A$, $\text{rbd}(A) = \{x : x_1^2 + x_2^2 = 1\}$ regardless of the space in which A is embedded.

10. The Separation Theorem

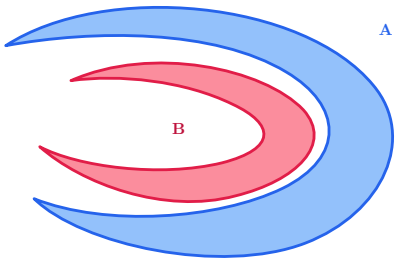
We will now prove one of the most fundamental properties of convex sets.

Definition 39 C and D are separable if there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x \leq b$ for all $x \in C$ and $a^T x \geq b$ for all $x \in D$.

Similarly, C and D are strictly separable if there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x < b$ for all $x \in C$ and $a^T x > b$ for all $x \in D$.

Theorem 40 Non-empty convex sets C and D can be separated if $\text{relint}(C) \cap \text{relint}(D) = \varnothing$.

This theorem in fact holds with "iff" if we strengthen the definition of separability so that at least one of C, D is not contained in the set $\{x : a^\top x = b\}$. Why do we need this additional condition? Note relint of a point is itself. Yet, as per the original definition, a point is separable from itself.



As seen above, convexity of *both* sets is necessary for this result. (Imagine B as an ellipse.)

This is a pretty fundamental result about convex sets. We will state a couple of corollaries that are convenient and then prove a weakened version.

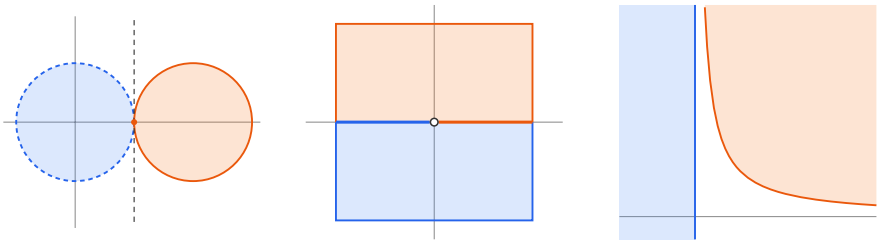
Corollary 40.1 Non-empty closed convex sets C and D such that $C \cap D = \varnothing$ are separable.

Corollary 40.2 Any non-empty convex set is separable from a point on its boundary.

Theorem 41 If C and D are closed convex non-empty sets such that $C \cap D = \varnothing$ and at least one of C, D is bounded, then there exist $a \neq 0 \in \mathbb{R}^n$, $b \in \mathbb{R}$ such that $a^T x < b$ for all $x \in C$ and $a^T x > b$ for all $x \in D$.

We will prove this result. This is weaker than the separation theorem in many ways, but also implies strict separation.

Let us examine why each condition is required.



Example. Take C to be the open unit diesk centered at $(0,0)$ and D to be the closed unit disk centered at $(2,0)$. These are convex, bounded, and $C \cap D = \varnothing$. Yet they are not strictly separable.

You might say that maybe being closed is not required if we weaken strict separability to strict inequality only on one side. That does not work either.

Example. In two dimensions, take $C = (\{x : x_2 < 0\} \cup \{x : x_2 = 0, x_1 < 0\}) \cap \{x : \|x\|_\infty \leq 1\}$, $D = (\{x : x_2 > 0\} \cup \{x : x_2 = 0, x_1 > 0\}) \cap \{x : \|x\|_\infty \leq 1\}$, which are convex and bounded, with $C \cap D = \varnothing$. Yet they are not strictly separable (even under the one-sided definition).

Similarly, one of the sets being bounded is necessary.

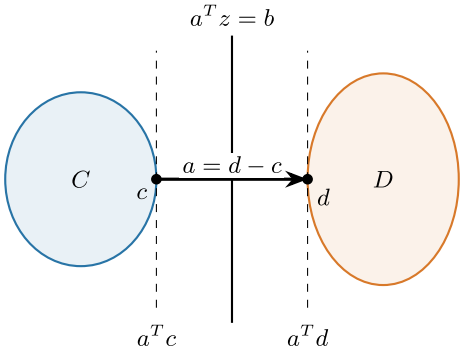
Example. In two dimensions, take $C = \{x : x_1 \leq 0\}$, $D = \{x : x_1 x_2 \geq 1, x_1 \geq 0\}$, which are closed and convex, with $C \cap D = \varnothing$. Yet they are not strictly separable.

Proof. Generically, say C is bounded. Consider $f(x) = \inf_{d \in D} \|x - d\|$ over $x \in C$. Now C is bounded, f is continuous (Why?). Hence, $\min_{x \in C} f(x)$ is achieved at some $c \in C$; let's fix this c.

Now, consider $f(x) = \|x - c\|$ over $x \in D$. Now f is continuous with bounded sub-level sets, so min of $f(x)$ over $x \in D$ is attained. Hence let $(c, d) = \operatorname{argmin}_{c \in C, d \in D} \|c - d\|$.

The idea hereafter is simple. The bisector of the two points is the hyperplane we are looking for. Take $a = d - c \neq 0$ since else $C \cap D \neq \varnothing$ and $b = \frac{\|d\|^2 - \|c\|^2}{2}$. By construction, $a^\top \left(\frac{c+d}{2}\right) = b$, or equivalently, $\frac{c+d}{2}$ lies on the hyperplane.

We claim $a^\top x < b$ for all $x \in C$ and $a^\top x > b$ for all $x \in D$. This might seem obvious, but I encourage you to try to complete the proof before reading further.



Choose any $d' \in D$. Since $f(x) = \frac{1}{2}\|x - c\|^2$ is minimized at d, from Proposition 27, we know that

$$\langle \nabla f(d), d' - d \rangle = \langle d - c, d' - d \rangle \geq 0, \quad \forall d' \in D.$$

Using this, we get

$$\begin{aligned} a^\top d' &= (d-c)^\top d' \geq (d-c)^\top d \\ &= (d-c)^\top \left(\frac{d+c}{2} + \frac{d-c}{2} \right) = b + \frac{\|c-d\|^2}{2} > b. \end{aligned}$$

This works (anti)symmetrically for any $c' \in C$. □

11. Equivalent Conditions for Convexity

Proposition 42 A differentiable function f is convex on a convex set K if and only if, for all $x, y \in K$, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Proof. $\Leftarrow$ Consider any $x, y \in K$, and $\lambda x + (1-\lambda)y \in K$ for some $\lambda \in [0,1]$. Now

$$f(x) \geq f(\lambda x + (1-\lambda)y) + (1-\lambda)\langle \nabla f(\lambda x + (1-\lambda)y), x - y \rangle,$$

$$f(y) \geq f(\lambda x + (1-\lambda)y) + \lambda \langle \nabla f(\lambda x + (1-\lambda)y), y - x \rangle.$$

Add (1)$\times\lambda$ + (2)$\times(1-\lambda)$ to get $\lambda f(x) + (1-\lambda) f(y) \geq f(\lambda x + (1-\lambda)y)$.

$\Rightarrow$ We know $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$, and hence

$$\langle \nabla f(y), x - y \rangle = \lim_{\lambda \to 0^+} \frac{f(y + \lambda(x-y)) - f(y)}{\lambda} \leq f(x) - f(y).$$

□

Proposition 43 A differentiable function f on a convex set K is convex if and only if, for all $x, y \in K$, we have

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0.$$

Proof. $\Rightarrow$ We know

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle,$$

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle.$$

Add these.

$\Leftarrow$ Using the fundamental theorem of calculus, we have

$$
\begin{aligned}
f(y) &= f(x) + \int_0^1 \langle \nabla f(x + \lambda(y-x)), y - x \rangle \, \mathrm{d}\lambda \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \langle \nabla f(x + \lambda(y-x)) - \nabla f(x), y - x \rangle \, \mathrm{d}\lambda \\
&\geq f(x) + \langle \nabla f(x), y - x \rangle.
\end{aligned}
$$

□

Proposition 44 A twice-differentiable function f is convex on a convex set K with non-empty interior if and only if, $\forall x \in \operatorname{int}(K)$, we have

$$\nabla^2 f(x) \succeq 0.$$

Proof. $\Leftarrow$ Note that

$$
\begin{aligned}
f(y) &= f(x) + \int_0^1 \langle \nabla f(x + \lambda(y-x)), y - x \rangle \, \mathrm{d}\lambda \\
&= f(x) + \langle \nabla f(x), y - x \rangle + \int_0^1 \int_0^\lambda (y-x)^\top \nabla^2 f(x + \mu(y-x))(y-x) \, \mathrm{d}\mu \, \mathrm{d}\lambda \\
&\geq f(x) + \langle \nabla f(x), y - x \rangle.
\end{aligned}
$$

$\Rightarrow$ Consider any $x \in \operatorname{int} K$; then there exists $\overline{\varepsilon} > 0$ such that $\forall 0 < \varepsilon < \overline{\varepsilon}, x + \varepsilon d \in K$ for any unit vector d. Now, let $y = x + \varepsilon d$.

$$
\begin{aligned}
0 &\leq \frac{\langle \nabla f(y) - \nabla f(x), y - x \rangle}{\varepsilon^2} \\
&= \frac{1}{\varepsilon} \langle \nabla f(y) - \nabla f(x), d \rangle \\
&= \frac{1}{\varepsilon} \int_0^\varepsilon d^\top \nabla^2 f(x + \mu d) d \, \mathrm{d}\mu.
\end{aligned}
$$

Take $\lim_{\varepsilon \to 0^+}$. Since this holds for all d, we are done. (Why?) □

11.1. First-Order Local-to-Global for Convex Functions

Finally, our efforts have paid off. We have the following local condition for convexity that can be cheaply checked and characterizes global optimality.

Corollary 44.1 Let f be a differentiable convex function on K. Then, $\overline{x}$ is a global minimum on K if and only if

$$\langle \nabla f(\overline{x}), x - \overline{x} \rangle \geq 0, \quad \forall x \in K.$$

This follows by combining Proposition 27 and Proposition 42.

Also, conveniently, for unconstrained optimization, or more generally when $\overline{x} \in \text{int}(K)$, this condition is equivalent to $\nabla f(\overline{x}) = 0$.

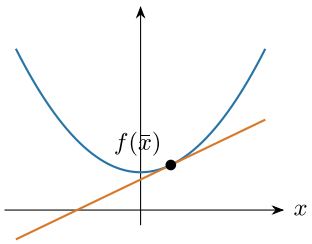
11.2. Subgradients and Their Existence

Convex univariate functions are differentiable everywhere except at countably many points. However, functions of the latter form might not be convex at all, e.g. a step function.

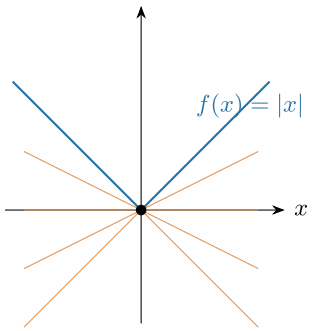
In convex analysis, there is an alternative way to make this concrete with a global definition, that generalizes gradients.

Definition 45 g is a subgradient of f at x on K if for all $y \in K$, $f(y) \geq f(x) + g^{T(y-x)}$.

The set of all subgradients of f at x on K is called the subdifferential $\partial f(x)$.



One supporting slope
$g = f'(\bar{x})$



Many supporting slopes
$\partial f(0) = [-1, 1]$

Theorem 46 Let f be a function on a convex set K. It is true that:
1. If for all $x \in K$ $\partial f(x) \neq \varnothing$, then f is convex, and
2. If f is convex, then, for all $x \in \text{int } K$, $\partial f(x) \neq \varnothing$.

Proof. (1) Same proof as $\Leftarrow$ for Proposition 42.

(2) Take any $x \in \text{int}(K)$. Then $(f(x), x)$ is on the boundary of $\text{epi}(f)$. Thus, $\exists(\alpha, \beta) \neq 0$ such that

$$\alpha t + \beta^\top y \geq \alpha f(x) + \beta^\top x \quad \forall (t, y) \in \text{epi}(f).$$

We claim $\alpha > 0$. If so, taking $t = f(y)$, $f(y) \geq f(x) + \left(-\frac{\beta}{\alpha}\right)^\top (y - x)$, and $\partial f(x) \neq \varnothing$.

To finish up: First, $\alpha \geq 0$ follows from taking $t \to \infty$, which is compatible with the epigraph. Finally, there exists $\varepsilon > 0$ such that $y = x + \varepsilon d \in K$ for all unit vectors d. Then, if $\alpha = 0$, note that $\beta^\top (y - x) = \beta^\top d\varepsilon \geq 0$ for all unit vectors d which can only happen if $\beta = 0$, but both cannot be zero simultaneously. □

Exercise. Prove that:
1. Non-negative sums of convex functions are convex.
2. $f(Ax + b)$ is convex for any convex f.
3. $\max_y f(x, y)$ is convex if for all y, $x \mapsto f(x, y)$ is convex.
4. $g(f(x))$ is convex if f is convex and g is convex nondecreasing.

5. $\min_y f(x, y)$ is convex if f is jointly convex in (x, y).
6. $g(x, y) = yf\left(\frac{x}{y}\right)$ on $\{(x, y) : y > 0\}$ is convex if f is convex on $\mathbb{R}^n$.

47-851: Modern Convex Optimization Lecture 3

Conic Programming and Duality

Lecturer: Karan Singh

This marks the beginning of the second unit of the course. So far, we have treated convexity at an abstract (generic) level. In this unit, we will look at structured optimization problems, where we can peek at the internals and try to make more fine-grained statements. We will begin by trying to generalize the best-understood family of convex programs: linear programs.

Contents

12. From Linear Programs to Conic Programs 24
13. A Hierarchy of Conic Programs 26
    13.1. The General Form of Second-Order Conic Programs 27
    13.2. The General Form of Semi-Definite Programs 27
    13.3. Back to The Hierarchy 28
14. Deriving SDP Duals 29
    14.1. The Weak Minimax Theorem 29
    14.2. Weak Duality for SDPs 29
15. Application: Approximation Algorithms for MaxCut 31
    15.1. Cuts and Laplacians 31
    15.2. Nesterov's Approximation for BQPs 31
16. Application: Stabilizability of Linear Dynamical Systems 33
    16.1. Lyapunov Stability 33
    16.2. Moving onto Stabilizability 35
17. Conic Duality 36
    17.1. Dual Cones and Weak Conic Duality 36
    17.2. Proving Strong Conic Duality 38
18. The S-Lemma 39
    18.1. The Homogenous Case 40
    18.2. Finishing up the General Case 41

12. From Linear Programs to Conic Programs

Let's recall linear programming duality. We'll build up to conic duality in the next two lectures, and this will serve as a good comparison.

$$\begin{aligned} &\textbf{Dual: } \min c^\top x && \textbf{Primal: } \max \quad b^\top y \\ &\text{subject to } Ax \geq b. && \text{subject to } A^\top y = c, \\ & && \quad y \geq 0. \end{aligned}$$

Observation: For any primal feasible y and dual feasible x, $c^\top x \geq b^\top y$. This is called weak duality.

Theorem 47 (Strong LP Duality) These conditions are equivalent:
1. Dual is feasible & bounded below.
2. Primal is feasible & bounded above.
3. Dual is solvable.
4. Primal is solvable.
5. Both primal & dual are feasible.

Whenever they hold, both primal and dual are equal in value.

Let us try to abstract what $\geq$ in the dual means. Of course, we know that for LPs, this corresponds to a cooridnate-wise comparison of vectors, but our goal is to isolate the essential structure.

For all $a, b, c, d \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}_{\geq 0}$, it is true that
1. $a \geq a$,
2. $a \geq b$ & $b \geq c \Rightarrow a \geq c$,
3. $a \geq b$ & $b \geq a \Rightarrow a = b$,
4. $a \geq b$ & $c \geq d \Rightarrow a + c \geq b + d$,
5. $a \geq b \Rightarrow \lambda a \geq \lambda b$.

Definition 48 A set K in a real vector space is a cone if for all $\lambda \geq 0$ and $x \in K$, $\lambda x \in K$.

Exercise. Show that a cone is convex if and only if it is closed under addition.

Similarly, in the primal, $y \geq 0 \equiv y \in K = \mathbb{R}^n_{\geq 0}$, where
1. K is a cone,
2. K is pointed, i.e., $a \in K$ & $-a \in K \Rightarrow a = 0$,
3. K is closed under addition.

Thankfully, with the interpretation that $a \geq b \equiv a - b \in K$, these two generalizations (or abstractions) can be seen to be equivalent. For example:
1. $a \geq a$ follows from K being a cone.
2. $a \geq b$ & $b \geq c \Rightarrow a \geq c$ follows from K being a cone and closed under addition.
3. $a \geq b$ & $b \geq a \Rightarrow a = b$ follows from K being pointed.
4. $a \geq b$ & $c \geq d \Rightarrow a + c \geq b + d$ follows from K being closed under addition.
5. $a \geq b \Rightarrow \lambda a \geq \lambda b$ follows from K being a cone.

Conic programs generalize LPs from $\mathbb{R}^n_{\geq 0}$ to arbitrary, closed convex pointed cones with non-empty interiors. (More on the non-empty interior condition in a couple of lectures.) A general conic program involves linear optimization over a feasible set, which is an intersection of an affine space and such a cone.

$$\begin{aligned} &\max c^\top x \\ &\text{subject to } Ax = b, \\ &\qquad\qquad x \in K. \end{aligned}$$

A general conic program.

13. A Hierarchy of Conic Programs

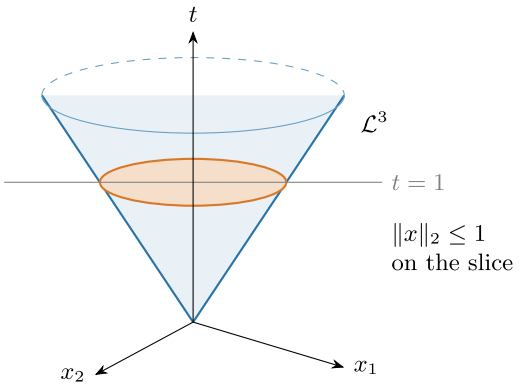
Conic programs can in fact represent any convex program. Let's do this in two steps. First, we can always make the objective linear.

$$\min_{x \in K} f(x) \equiv \min_{(t,x) \in \text{ epi}(f)} t.$$

To finish, we need to make the set a cone. Consider the cone $C(K) = \{\lambda(1, x) : \lambda \geq 0, x \in K\}$.

$$\begin{aligned} \min_{x \in K} c^\top x \equiv \min\ & c^\top x \\ & \text{subject to } t = 1 \\ & \text{and } \ (t, x) \in C(K). \end{aligned}$$

So to specialize, we will look at *specific* cones. Examples of admissible cones:
1. $\mathbb{R}^n_{\geq 0}$ gives us linear programs.
2. The Lorentz cone $\mathcal{L}^{n+1} = \{(t, x) : t \geq \|x\|_2\}$ gives us SOCPs (second-order conic programs).
3. $S^{n \times n}_+ = \{S : S \succeq 0\}$ gives us SDPs (semi-definite programs).



Let us define a bunch of optimization problems.

$$\textbf{QPs: } \min x^\top Q x + q^\top x + d \text{ with } Ax \leq b.$$

$$\textbf{QCQPs: } \min x^\top Q x + q^\top x + d \text{ with } x^\top Q_i x + q_i^\top x + d_i \leq 0 \text{ for all } i \in [m].$$

$$\textbf{SOCPs: } \min f^\top x \text{ with } \|A_i x + b_i\|_2 \leq c_i^\top x + d_i \text{ for all } i \in [m].$$

$$\textbf{SDPs: } \min \operatorname{tr}(CX) \text{ with } \operatorname{tr}(A_i X) = b_i \text{ for all } i \in [m],\ X \succeq 0.$$

SOCPs and SDPs are always convex programs. But that is not true for Quadratic Programs (QPs) and Quadratically Constrained Quadratic Programs (QCQPs). More carefully, Convex QPs are those where $Q \succeq 0$, and similarly Convex QCQPs are those with $Q, Q_1, ...Q_m \succeq 0$.

Note that below $\langle A, B \rangle = \operatorname{tr}(A^\top B) = \sum_{i,j=1}^n A_{ij} B_{ij}$ and $\langle A, A \rangle = \|A\|_F^2 = \sum_{i,j=1}^n A_{ij}^2$.

13.1. The General Form of Second-Order Conic Programs

Let's take a second to appreciate why SOCPs as described can be written as the intersection of an affine space and the Lorentz cone. Clearly:

$$\|Ax + b\|_2 \leq cx + d \equiv (cx + d, Ax + b) \in \mathcal{L}^{n+1}.$$

This is an affine map. Recall any (linear) subspace can be expressed as the column-space of a matrix (the generator view) and as the null-space of another (the constraint view). Thus, by the fundamental theorem of linear algebra, there exist E, f such that

$$\{(cx + d, Ax + b) : x \in \mathbb{R}^n\} = \{y : Ey = f\}.$$

As a concrete example, consider that

$$\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} x : x \in \mathbb{R} \right\} \equiv \left\{ y : \begin{pmatrix} 2 \\ -1 \end{pmatrix}^\top y = 0 \right\}.$$

Essentially, the null-space vectors are all the orthogonal vectors the generators leave out. Thus, this equivalence can also be algorithmically carried out by Gram-Schmidt. Geometrically, we are appealing to the fact that all hyperplanes can be described by an anchoring point and a bunch of vectors to create the span about the anchoring point, and equivalently as the anchoring point together with a normal vector to the hyperplane.

13.2. The General Form of Semi-Definite Programs

Let's talk about SDPs. SDPs are tricky.

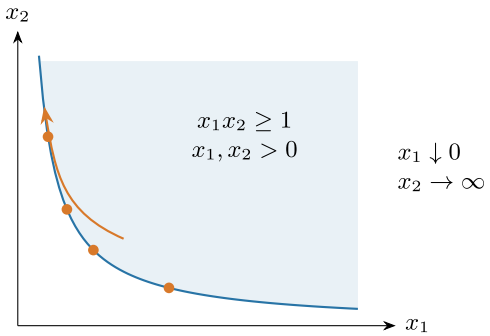
Why not impose linear constraints as $AX = B$ instead? Not all linear constraints can be written this way, e.g., $X_{11} = 0$, which is same as $0 = e_1^\top X e_1 = \operatorname{tr}(X e_1 e_1^\top)$.

Example. SDPs can have infinitely many extreme points.

$$S = \{(x, y) : x^2 + y^2 \leq 1\} = \left\{ (x, y) : \begin{pmatrix} 1 + x & y \\ y & 1 - x \end{pmatrix} \succeq 0 \right\}.$$

Example. For SDPs, feasibility & bounded below $\not\Rightarrow$ solvable.

$$\min x_1 \quad \text{subject to} \quad \begin{pmatrix} x_1 & 1 \\ 1 & x_2 \end{pmatrix} \succeq 0.$$



13.3. Back to The Hierarchy

Theorem 49 LPs $\subseteq$ Convex QPs $\subseteq$ Convex QCQPs $\subseteq$ SOCPs $\subseteq$ SDPs.

Exercise. Write $\max c^\top x$ subject to $Ax \geq b$ as directly an SDP.

A useful result in proving the above theorem is next.

Lemma 50 (Schur Complements) Let $A \succ 0$. Then

$$X = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} \succeq 0 \text{ if and only if } C - B^\top A^{-1} B \succeq 0.$$

Here $C - B^\top A^{-1} B$ is called the Schur complement of A in X.

Proof. Verify the following factorization:

$$X = \begin{pmatrix} I & 0 \\ B^\top A^{-1} & I \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & C - B^\top A^{-1} B \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & I \end{pmatrix} = P^\top S P.$$

The first and last matrices are transposes of the other and invertible. In fact, for any triangular matrix, its eigenvalues can be read off the diagonal. (Why?)

Note that $x^\top X x = (Px)^\top S (Px)$ and $y^\top S y = (P^{-1}y)^\top X (P^{-1}y)$. Hence, $X \succeq 0$ if and only if the middle matrix is PSD if and only if $C - B^\top A^{-1} B \succeq 0$. □

In fact, such matrices S and X are said to be *congruent*. A well-know fact, although not common in a first linear-algebra course, is that P and X have the same *inertia*, where inertia of a matrix is a 3-tuple composed of the number of positive, negative, and zero eigenvalues. Thus

$$\text{inertia}(X) = \text{inertia}(A) + \text{inertia}(C - B^\top A^{-1} B),$$

of which we are proving a special case.

A related notion that I sometimes confuse this with is *similar* matrices. If S and X were similar, then it would hold that $X = P^{-1} S P$, and a much stronger conclusion would follow that all eigenvalues are preserved in similarity transformation. (Correct me if this happens!)

We are now ready to prove the main theorem.

Proof. The first two inclusions are straightforward. We can take $Q = 0$ and $Q_i = 0$, respectively.

(Convex QCQPs $\subseteq$ SOCPs) For any $Q = \sum_i \lambda_i u_i u_i^\top \succeq 0$, we can define $Q^{\frac{1}{2}} = \sum_i \sqrt{\lambda_i} u_i u_i^\top \succeq 0$. Thus completing the square

$$x^\top Q x + 2q^\top x + d \leq 0 \equiv \left\| Q^{\frac{1}{2}} x + Q^{-\frac{1}{2}} q \right\|_2 \leq \sqrt{q^\top Q q - d},$$

but this only works if $Q \succ 0$ (strictly).

(SOCPs $\subseteq$ SDPs) Using the Schur Complement Lemma, $\|Ax + b\| \leq cx + d$ if and only if

$$c^\top x + d - \frac{\|Ax+b\|^2}{c^\top x + d} \geq 0 \equiv \begin{pmatrix} (cx+d)I & Ax+b \\ (Ax+b)^\top & cx+d \end{pmatrix} \succeq 0.$$

This is clear if $c^\top x + d > 0$. Incidentally the end-to-end conclusion also works when $c^\top x + d = 0$. (Why?) Thus, SOCPs $\subseteq$ SDPs. □

Exercise. Fix the above proof so that QCCPs can be written as SOCPs even when $Q \succeq 0$.

Hint: There are many ways to do this. For example, via SVDs, though not directly; not clear why we can tolerate a $\sqrt{\text{linear function}}$ term on the RHS. One idea is to make use of the following observation:

$$\left\| \begin{pmatrix} x \\ y-1 \end{pmatrix} \right\|_2 \leq y+1 \equiv x^2 \leq 4y$$

14. Deriving SDP Duals

Here we will find a way to derive SDP duals. The proof of strong duality will come next week.

14.1. The Weak Minimax Theorem

All dualities start with the following elementary lemma. In words, it says without assuming anything about the function f, the second mover has the ultimate say. If min/max do not exist, interpret these as inf/sup for this section.

Lemma 51 $\min_{x\in X} \max_{y\in Y} f(x,y) \geq \max_{y\in Y} \min_{x\in X} f(x,y)$.

Proof. Clearly, for all $x \in X$, $y \in Y$, $f(x,y) \geq \min_{x\in X} f(x,y)$. Also, $\max_{y\in y} g(y) \geq \max_{y\in y} h(x)$ is a valid conclusion if $g(y) \geq h(y)$ for all $y \in y$. Using this, we get for all $x \in X$ that

$$\max_{y\in Y} f(x,y) \geq \max_{y\in Y} \min_{x\in X} f(x,y).$$

Because the last inequality holds for all $x \in X$, we also have that

$$\min_{x\in X} \max_{y\in Y} f(x,y) \geq \max_{y\in Y} \min_{x\in X} f(x,y).$$

□

14.2. Weak Duality for SDPs

Let's try to apply this to SDPs. To begin, we have

$$\begin{aligned} &\min_X \langle C, X\rangle \ \text{ subject to } \ \langle A_i, X\rangle = b_i \forall i \in [m] \ \text{ and } \ X \succeq 0 \\ &= \min_{X\succeq 0} \max_{y\in\mathbb{R}^m} \langle C, X\rangle + \sum_{i=1}^m y_i(b_i - \langle A_i, X\rangle) \end{aligned}$$

This works because if there exists some $i \in [m]$ such that $b_i > \langle A_i, X\rangle$, then $y_i \to \infty$ making a minimization program approach $+\infty$ in value, which by convention is same as infeasibility.

Similarly, if $b_i < \langle A_i, X\rangle$, then $y_i \to -\infty$ making a minimization program again approach $+\infty$ in value. Now, we use by minimax result as

$$\begin{aligned}
&\min_X \langle C, X\rangle \;\text{ subject to }\; \langle A_i, X\rangle = b_i \forall i \in [m] \;\text{ and }\; X \succeq 0 \\
&= \min_{X\succeq 0} \max_{y\in\mathbb{R}^m} \langle C, X\rangle + \sum_{i=1}^m y_i(b_i - \langle A_i, X\rangle) \\
&= \min_{X\succeq 0} \max_{y\in\mathbb{R}^m} b^\top y + \left\langle C - \sum_{i=1}^m y_i A_i, X \right\rangle \\
&\geq \max_{y\in\mathbb{R}^m} \min_{X\succeq 0} b^\top y + \left\langle C - \sum_{i=1}^m y_i A_i, X \right\rangle.
\end{aligned}$$

To simplify the inner optimization problem, note that if $C \not\succeq \sum_{i=1}^m y_i A_i$, there exists an eigenvectors u with eigenvector λ such that $u^\top\big(C - \sum_{i=1}^m y_i A_i\big)u = \lambda < 0$. So, we can take $X = \alpha uu^\top$ with $\alpha \to +\infty$, so that $\left\langle C - \sum_{i=1}^m y_i A_i, X\right\rangle = \alpha u^\top\big(C - \sum_{i=1}^m y_i A_i\big)u \to -\infty$, which is the same as infeasibility for a maximization program. Also if $C \succeq \sum_{i=1}^m y_i A_i$ then for all $X \succeq 0$, we have that $\left\langle C - \sum_{i=1}^m y_i A_i, X\right\rangle \geq 0$, in which case the inner minimizer might as well choose $X = 0$.

Lemma 52 $A \succeq 0$, $B \succeq 0 \Rightarrow \langle A, B\rangle \geq 0$.

Proof. Let the spectral decoposition of A be $A = \sum_i \lambda_i u_i u_i^\top$. Here $A \succeq 0 \Rightarrow \lambda_i \geq 0$. Hence

$$\begin{aligned}
\langle A, B\rangle &= \sum_i \lambda_i \langle u_i u_i^\top, B\rangle \\
&= \sum_i \lambda_i u_i^\top B u_i \geq 0.
\end{aligned}$$

□

Using all this, we get

$$\begin{aligned}
&\min_X \langle C, X\rangle \;\text{ subject to }\; \langle A_i, X\rangle = b_i \forall i \in [m] \;\text{ and }\; X \succeq 0 \\
&\geq \max_{y\in\mathbb{R}^m} \min_{X\succeq 0} b^\top y + \left\langle C - \sum_{i=1}^m y_i A_i, X \right\rangle \\
&= \max_{y\in\mathbb{R}^m} \begin{cases} b^\top y \;\text{ if }\; C \succeq \sum_{i=1}^m y_i A_i \\ -\infty \text{ otherwise} \end{cases} \\
&= \max_{y\in\mathbb{R}^m} b^\top y \;\text{ subject to }\; C \succeq \sum_{i=1}^m y_i A_i.
\end{aligned}$$

This is the dual we sought. An inequality of the form $A_0 + \sum_{i=1}^m y_i A_i \succeq 0$ with variables y_i is called a *linear matrix inequality*. Again, it might not obvious at first sight if this dual is a SDP too. However, the LMI constraint is a PSD constraint on an affine map, and hence representable as intersection of an affine space and the PSD cone. Hence, the dual is a SDP too.

By the minimax claim, the only used inequality was a valid one. Hence, we have weak duality by construction.

Lemma 53 For any primal feasible X, dual feasible y, we have $\langle C, X\rangle \geq b^\top y$.

15. Application: Approximation Algorithms for MaxCut

In the next couple of sections, we will work through two applications of SDPs, one in the discrete math world and another in the continuous world.

15.1. Cuts and Laplacians

A cut of a graph is a binary partition of the vertices, and its size is measured by the number of edges crossing across the partition. A canonical NP-hard problem asks for the maximum cut.

$$\max_{S\subseteq V} |(S \times (V-S)) \cap E|.$$

For a graph $G = (V, E)$, the Laplacian $L \in \mathbb{R}^{n\times n}$ is as follows $L_{ii} = \deg(v_i)$ for all $i \in [n]$ and for all $i \neq j$, $L_{ij} = -\mathbf{1}\{(v_i, v_j) \in E\}$. To see why this is useful, observe that

$$x^\top L x = \sum_{i=1}^n \deg(i) x_i^2 - 2\sum_{(i,j)\in E} x_i x_j = \sum_{(i,j)\in E} (x_i - x_j)^2.$$

This means that $L \succeq 0$. Also, for any $S \subseteq V$, we if let $x_S \in \mathbb{R}^n$ be its (signed) characteristic vector, that is, $x_i = \mathbf{1}\{x \in S\} - \mathbf{1}\{x \notin S\}$, then $\frac{1}{4} x_S^\top L x_S = \mathrm{Cut}(S)$.

15.2. Nesterov's Approximation for BQPs

In fact, consider any binary quadratic program with PSD cost matrix; MaxCut is a special case. Because any rank-1 PSD matrix can be expressed as $vv^\top$, we have

$$\begin{aligned}
\mathrm{BQP} &\equiv \max_{x_i^2=1, \forall i\in[n]} x^\top A x \\
&= \max_{x_i^2=1, \forall i\in[n]} \langle A, xx^\top\rangle \\
&= \max\langle A, X\rangle \ \text{ subject to } \ X_{ii} = 1, \forall i \in [n] \ \text{ and } \ X \succeq 0, X \ \text{ is rank-1.}
\end{aligned}$$

Define $\mathrm{SDP} \equiv \max\langle A, X\rangle$ with $X_{ii} = 1$ for all $i \in [n]$, $X \succeq 0$. Clearly $\mathrm{SDP} \geq \mathrm{BQP}$.

Theorem 54 (Nesterov) $\mathrm{BQP} \geq \left(\frac{2}{\pi}\right) \mathrm{SDP}$ for BQP with $A \succeq 0$.

In fact, the proof of this theorem provides a $\frac{2}{\pi} \approx 0.636$ approximation algorithm for such BQPs. For MaxCut, the same SDP can also be used to produce a ≈ 0.878 approximation (which is unimprovable up to UGC). But the above result is more general; we don't use the fact that A is a Laplacian. Also, random cuts, greedy methods, LP relaxations all produce $\frac{1}{2}$ approximation. So, this is not unimpressive.

Let us prove the following helper results first.

Lemma 55 $A, B \succeq 0 \Rightarrow A \odot B \succeq 0$, where $\odot$ is the element-wise product.

Proof. There exist $v_1, ..., v_n$ such for all i, j, we have $A_{ij} = v_i^\top v_j$. Similarly, there exist $u_1, ..., u_n$ such for all i, j, we have $B_{ij} = u_i^\top u_j$. Consequently, we have

$$\begin{aligned}(A \odot B)_{ij} &= \big(v_i^\top v_j\big)\big(u_i^\top u_j\big) \\ &= \sum_{k,\ell} v_{ik} v_{jk} u_{i\ell} u_{j\ell} \\ &= \Big\langle u_i v_i^\top, u_j v_j^\top \Big\rangle.\end{aligned}$$

We are essentially vectorizing the rank-1 matrices. Thus $A \odot B \succeq 0$. □

Corollary 55.1 $A \succeq 0 \Rightarrow A^{\odot n} \succeq 0$ for all $n \in \mathbb{Z}_{\geq 0}$, where $\odot$ indicates elementwise powers.

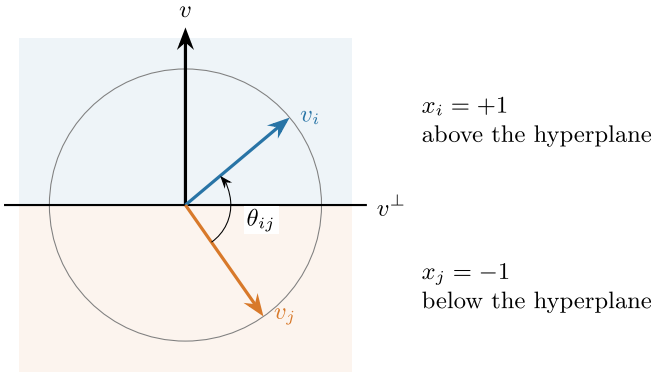
Lemma 56 $\sin^{-1}(X) \succeq X$ for all $X \succeq 0$, where $\sin^{-1}$ is applied element-wise.

Proof. Consider the power series which converges to $\sin^{-1}(x)$ uniformly on $[-1, 1]$, that is,

$$\begin{aligned}\sin^{-1}(x) &= x + \frac{x^3}{2 \times 3} + (1 \times 3)\frac{x^5}{2 \times 4 \times 5} + (1 \times 3 \times 5)\frac{x^7}{2 \times 4 \times 6 \times 7} + ... \\ &= x + \sum_{k \geq 2} a_k x^k\end{aligned}$$

for some $a_k \geq 0$. Using the last corollary, $\sin^{-1}(X) \succeq X$. □

Proof. Consider any SDP-feasible $X \in \mathbb{R}^{n \times n}$. There exist $v_1, ..., v_n \in \mathbb{R}^n$ such that $X_{ij} = v_i^\top v_j$. Since $X_{ii} = 1$, v_i's are unit vectors. Take a random unit vector $v \sim \text{Unif}(S^{n-1})$. Construct a vector x with $x_i = \text{sign}(\langle v, v_i \rangle)$, where $\text{sign}(z) = \mathbf{1}\{z \geq 0\} - \mathbf{1}\{z < 0\}$. Clearly, x is BQP feasible.



$$\begin{aligned}
\mathbb{E}[x_i x_j] &= \Pr\big(\operatorname{sign}(\langle v, v_i\rangle) = \operatorname{sign}(\langle v, v_j\rangle)\big) - \Pr\big(\operatorname{sign}(\langle v, v_i\rangle) \neq \operatorname{sign}(\langle v, v_j\rangle)\big) \\
&= 1 - 2\Pr\big(\operatorname{sign}(\langle v, v_i\rangle) \neq \operatorname{sign}(\langle v, v_j\rangle)\big) \\
&= 1 - \frac{4\cos^{-1}(\langle v_i, v_j\rangle)}{2\pi} \\
&= \frac{2}{\pi}\Big(\frac{\pi}{2} - \cos^{-1}(\langle v_i, v_j\rangle)\Big) \\
&= \frac{2}{\pi}\sin^{-1}(\langle v_i, v_j\rangle) = \frac{2}{\pi}\sin^{-1}(X_{ij}).
\end{aligned}$$

Thus, we arrive at

$$\text{BQP} \geq \mathbb{E}[x^\top A x] = \sum_{i,j} A_{ij}\mathbb{E}[x_i x_j] = \frac{2}{\pi}\langle A, \sin^{-1} X\rangle,$$

where $\sin^{-1}$ is applied elementwise.

To complete the proof, take X to be the SDP solution. (Why does this exist?) Then, we get

$$\begin{aligned}
\text{BQP} &\geq \frac{2}{\pi}\langle A, \sin^{-1}(X)\rangle \\
&= \frac{2}{\pi}(\langle A, \sin^{-1}(X) - X\rangle + \langle A, X\rangle) \\
&\geq \frac{2}{\pi}\ \text{SDP}.
\end{aligned}$$

□

16. Application: Stabilizability of Linear Dynamical Systems

Let's do another application, before we move onto conic duality.

16.1. Lyapunov Stability

A fundamental question about dynamical system $x_{t+1} = f(x_t)$ is if their long-term behaviors can be neatly categorized. As a concrete question, consider a linear dynamical system $x_{t+1} = Ax_t$. When does such a system always end up at 0 (i.e., $x_t \to 0$ as $t \to \infty$) regardless of initialization? (So for every x_0.) Such systems are called stable.

Definition 57 A linear dynamical system with transition matrix A is stable if $\forall x_0 \in \mathbb{R}^n$, $x_t = A^t x_0 \to 0$ as $t \to \infty$.

Consider a diagonalizable matrix $A = U\Sigma U^{-1}$. Here, U is the matrix of eigenvectors concatenated together, and Σ is a diagonal matrix. Note that the eigenvectors may not be orthogonal, neither might the eigenvalues be real. Almost all matrices are diagonalizable. Any x can be written as $x = \sum_i c_i u_i$. Hence $A^t x = \sum_i \lambda_i^t c_i u_i$. From this we get:

Lemma 58 A is stable if and only if $\lim_{t\to\infty} A^t = 0$ if and only if $\rho(A) < 1$.

Here, the spectral radius $\rho(A) = \max_i |\lambda_i|$ is the largest modulus of any eigenvalue. Unlike the spectral (or operator) norm, $\|A\|_2 = \max_{\|x\|_2 \le 1} \|Ax\|_2$, $\rho(\cdot)$ is non-convex.

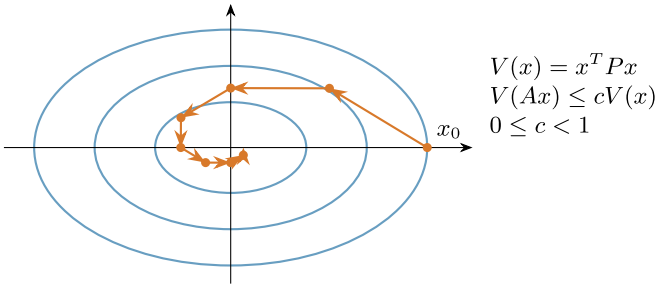
Example. Consider $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $\rho(A) = \rho(B) = 0$, $\|A\|_2 = \|B\|_2 = 1$.

Yet, $\rho\left(\frac{A+B}{2}\right) = \frac{1}{2}$, $\left\|\frac{A+B}{2}\right\|_2 = \frac{1}{2}$.

Stability is a property that holds over all trajectories, each defined with respect to a unique starting state. Lyapunov gave a criteria to certify the stability of a system without analyzing all its trajectories. The idea is to find a function $V(x)$ such

1. $V(x) > 0$ for all $x \neq 0$,
2. $V(0) = 0$,
3. $V(f(x)) \le V(x)$ for all x, with strict inequality for $x \neq 0$.

Under such conditions, it is clear $V(x_t)$ decreases asymptotically. The surprising bit is that he showed such a certificate always exists for *nice-enough* stable systems. Today, such analysis is called potential-based or Lyapunov analysis.



Lemma 59 A is stable if and only if there exists $P \succeq I$ such $P \succ A^\top P A$.

Proof. $\Leftarrow$ Take $V(x) = x^\top P x$ where $P \succeq I$, $cP \succeq A^\top P A$ for some $0 \le c < 1$.

Then $V(0) = 0$, $V(x) \ge \|x\|^2 > 0$ for all $x \neq 0$ and $V(Ax) = x^\top A^\top P A x \le cV(x)$. Therefore, $V(A^t x) \le c^t V(x) \to 0$ as $t \to \infty$, which also implies $x_t \to 0$ as $t \to \infty$.

$\Rightarrow$ Take $P = \sum_{t\ge 0} (A^t)^\top A^t$. Clearly, $P \succeq I$ as $x_0^\top P x_0 = \sum_{t\ge 0} \|x_t\|^2$. Also

$$A^\top P A = \sum_{t\ge 0} (A^{t+1})^\top A^{t+1} = P - I,$$

from which we have $P \succ A^\top P A$. The only bit left out is that P as defined has to exist. We use the following fact that follows from Jordan decomposition:

Fact: $\|A^t\| \le O(t^n \rho(A)^t)$.

Hence P is well-defined, as $\rho(A) < 1$. □

Note that $\{P : P \succeq I, cP \succeq A^\top P A\}$ is a SDP feasibility program; and one can do bisection on the guess of c in $[0, 1)$, to solve it arbitrarily well.

16.2. Moving onto Stabilizability

A more involved question is that of stabilizability. Consider a dynamical system with inputs: $x_{t+1} = Ax_t + Bu_t$. Many systems are not intrinsically stable, but can be stable under correct choice of control inputs. This is what stabilizability captures.

Definition 60 A linear dynamical system (A, B) with inputs is stabilizable if for all $x_0 \in \mathbb{R}^n$, there exist $u_1, u_2, ... \in \mathbb{R}^m$ such $x_t \to 0$ as $t \to \infty$.

The dynamical systems we have considered so far are Markovian; all information at time t is summarized in the state x_t. Using this and linearity of dynamics, one can gain an alternate characterization.

Observation: (A, B) is stabilizable if and only if there exists K such $\rho(A + BK) < 1$.

The basic idea is to use inputs of the form $u_t = Kx_t$ for all x_t. If so,

$$x_{t+1} = Ax_t + Bu_t = (A + BK)x_t = (A + BK)^t x_0.$$

Thus the dynamical system looks like one without inputs where $A_{\text{eff}} = A + BK$. Since such a K exists with $\rho(A + BK) < 1$ whenever (A, B) is stabilizable, can one find it efficiently?

Note that $\min_K \rho(A + BK)$ is a non-convex program.

Instead, let's try to use our lemma.

$$\begin{aligned}
&(A, B) \text{ is stabilizable} \\
\equiv\ &\exists K, \rho(A + BK) < 1 \\
\equiv\ &\exists K, P \succeq I, P \succ (A + BK)^\top P(A + BK) \\
\equiv\ &\exists K, P \succeq I, \begin{pmatrix} P & ((A + BK)P)^\top \\ (A + BK)P & P \end{pmatrix} \succ 0. \quad (1)
\end{aligned}$$

Why? We already know $P \succ 0$, thus by the Schur complement lemma, satisfaction of the above constraint is same as

$$P - (A + BK)PP^{-1}P(A + BK)^\top \succ 0.$$

Note this does not match what we had in the last step. But that's fine since what we had was same as $\rho((A + BK)^\top) < 1$, because $(A + BK)^\top$ and $A + BK$ have the same characteristic equation and hence eigenvalues. However we still don't have a SDP at this stage due to non-linearity produced by multiplication of K and P; both are variables.

$$\begin{aligned}
&(A, B) \text{ is stabilizable} \\
\equiv\ &\exists K \in \mathbb{R}^{n \times m}, P \succeq I, \begin{pmatrix} P & (AP + BL)^\top \\ AP + BL & P \end{pmatrix} \succ 0. \quad (2)
\end{aligned}$$

At first glance, this might seem like a relaxation, we used $L = KP$ to get an (almost) SDP feasibility program. But this is in fact a reformulation since one can go back (i.e. construct a feasible solution for the original program) by setting $K = LP^{-1}$.

For this program, once again perform bisection with $\succ$ constraints.

What we were able to accomplish in the last step (an exact reformulation) was a lucky coincidence. As an example, if we want to impose a box constraint on K, we can do so in (1) which was (nominally) non-convex anyways, but now an exact reformulation in (2) as an SDP is no longer possible. In fact, this problem is NP-hard.

17. Conic Duality

In this section, we will ultimately prove strong duality for conic programs with general cones. For this, we will define generalized inequalities as $x \underset{K}{\geq} y$ if $x - y \in K$, where K is a cone.

17.1. Dual Cones and Weak Conic Duality

Consider $\min c^\top x$ subject to $Ax \underset{K}{\geq} b$. Before we get to strong duality, it is worth noting that it's not clear how we even define a dual in the first place. For LPs, our derivation was based on the fact that non-negative combinations of vector inequalities always collapse to valid scalar inequalities. But this is not true for CPs.

Example. $\mathcal{L}^{2+1} = \{(t, x) : t \geq \|x\|_2\}$. Clearly, $(1, 0, -1) \underset{\mathcal{L}^{2+1}}{\geq} (0, 0, 0)$. But

$$(0, 0, 1)^\top (1, 0, -1) = -1 \not\geq 0 = (0, 0, 1)^\top (0, 0, 0).$$

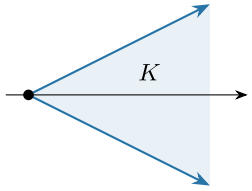
You might guess that the multiplier should come from K itself. This is also false.

Example. $K = \{(x, y) : |y| \leq 100x\}$. Then $(1, -100) \underset{K}{\geq} (0, 0)$. But $(1, 100) \in K$, yet

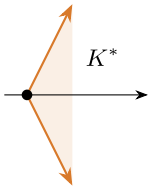
$$(1, 100)^\top (1, -100) = 1 - 100^2 \not\geq 0 = (1, 100)^\top (0, 0).$$

The true answer follows from a tautology. For all $y \in K_*$, $a \underset{K}{\geq} b \Rightarrow y^\top a \geq y^\top b$ for all $a, b \in K$.

Definition 61 The dual cone K_* of a set $K \subseteq \mathbb{R}^n$ such that $K_* = \{y : \forall x \in K, y^\top x \geq 0\}$.



$K = \{(u, w) : u \geq 2|w|\}$



$K^* = \{(a, b) : a \geq |b|/2\}$

Using the minimax inequality, once again we have

$$
\begin{aligned}
&\min_x c^\top x \;\text{ subject to }\; Ax \underset{K}{\geq} b && \text{(primal)}\\
&= \min_x \max_{y \in K_*} c^\top x + y^\top (b - Ax) && \text{(dubious)}\\
&\geq \max_{y \in K_*} \min_x c^\top x + y^\top (b - Ax)\\
&= \max_{y \in K_*} \min_x x^\top (c - A^\top y) + b^\top y\\
&= \max b^\top y \;\text{ subject to }\; A^\top y = c, y \in K_* && \text{(dual)}
\end{aligned}
$$

Exercise. Justify the *dubious* step in the derivation above. Specifically, if $Ax - b \notin K$, then why is $\min_{y \in K_*} y^\top (Ax - b) = -\infty$?

Hint: Use the strict separation theorem along with the fact that K is a closed convex cone.

For any primal-dual feasible pair (x, y), we get

$$c^\top x = y^\top A x = y^\top (Ax - b) + b^\top y \geq b^\top y.$$

Lemma 62 (Weak Duality) For primal feasible x, dual feasible y, we have $c^\top x \geq b^\top y$.

To progress towards strong duality, we need to understand the dual cone K_* a bit more.

Lemma 63 Let K be a non-empty set in $\mathbb{R}^n$. Then:
1. K_* is a closed convex cone.
2. $\text{int}(K) \neq \varnothing \Rightarrow K_*$ is pointed.
3. K is a closed convex pointed cone $\Rightarrow \text{int}(K_*) \neq \varnothing$.
4. K is a closed convex cone $\Rightarrow K_{**} = K$.

Exercise. Justify the *dubious* step again this time directly using part (4) of the above lemma.

Exercise. Prove part (3) of the above lemma.

Proof. (1) K_* can also we written as

$$K_* = \bigcap_{x \in K} \{y : y^\top x \geq 0\}.$$

is intersection of halfspaces (hence convex) and intersection of closed sets (hence closed). Cone is true from definition, because homogeneous inequalities are involved.

(2) Say K_* is not pointed. Then K_* contains an entire line passing through origin, say $\{\lambda a : \lambda \in \mathbb{R}\}$. Then $a^\top x \geq 0$ for all $x \in K$ and $(-a)^\top x \geq 0$ for all $x \in K$. Thus $a^\top x = 0$. But then K is flat, it's entirely contained in the maximal subspace orthogonal to a. Hence, $\text{int}(K) = \varnothing$.

(4) First $K \subseteq K_{**}$. Why? Pick any $x \in K$, clearly for all $y \in K_*$, $y^\top x \geq 0$. Thus $x \in K_{**}$. For the other direction, we'll invoke separation. Let's do the contrapositive. Say $x \notin K$. There exists a non-zero vector y such that $y^\top x < y^\top z$ for all $z \in K$. But then $z^\top y$ is lower bounded on K, hence $\min_{z \in K} z^\top y = 0$. Hence $y \in K_*$, and from $y^\top x < 0$, $x \notin K_{**}$. □

Corollary 63.1 K is a closed convex pointed cone with non-empty interior if and only if K_* is a closed convex pointed cone with non-empty interior.

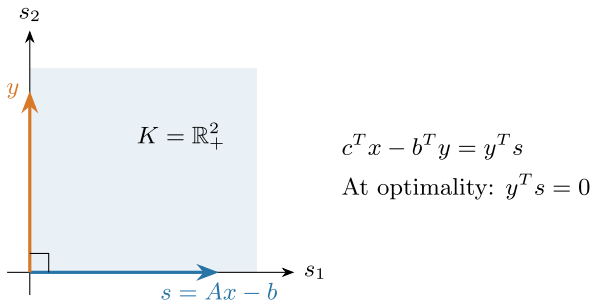
17.2. Proving Strong Conic Duality

Going forward, we will work with closed convex pointed cones with non-empty interiors. We will in fact see why the non-empty interior bit is important. For now, let us note that without it the dual is not pointed.

Theorem 64 If the primal is bounded below and strictly feasible (i.e. $\exists x, Ax \underset{K}{\geq} b$), then the dual is solvable and optimal values match.

By symmetry, if the dual is bounded above and strictly feasible ($y \underset{K_*}{\geq} 0$) then the primal is solvable and optimal values match.

Corollary 64.1 If one of the primal or the dual is bounded and strictly feasible, and (x, y) is primal-dual feasible pair, then (x, y) is primal-dual optimal if and only if $b^\top y = c^\top x$ if and only if $y^\top (Ax - b) = 0$.



Under given conditions, we know optimal values are equal. We also know from weak duality that for any pair of feasible solutions

$$b^\top y \leq D^* \leq P^* \leq c^\top x.$$

Thus attainment of optimal values at (x, y) for (P, D) happens if and only if $b^\top y = c^\top x$. From the proof of weak duality this is equivalent to $y^\top (Ax - b) = 0$, i.e., the only inequality is tight.

Proof of strong duality. Let P^* be the optimal value for the primal problem. We will construct a dual feasible y^* such $b^\top y^* \geq P^*$, via the separation theorem.

Let $V = \{Ax - b : c^\top x \leq P^*\}$. It is non-empty, because any affine/linear map of a halfspace can not be empty. Also $V \cap \operatorname{int}(K) = \varnothing$, otherwise given x such $c^\top x \leq P^*$ and $Ax \underset{K}{\geq} b$, one can strictly decrease the objective for a close enough feasible point, assuming $c \neq 0$.

As an aside, note that $c \neq 0$ holds generically. If not, $P^* = 0$. Take $y = 0$, $A^\top y = c$ holds and we have $0 = b^\top y \geq P^*$. Our work in this case is complete.

Now, by (actual) separation theorem, there exists $y \neq 0 \in \mathbb{R}^m$ such that

$$\sup_{x \in V} y^\top x \leq \inf_{x \in K} y^\top x.$$

So, $y^\top x$ is bounded below on K. But $\inf_{x \in K} y^\top x$ generically only take values $\{0, -\infty\}$. Thus

$$\sup_{x \in V} y^\top x \leq 0 \leq \inf_{x \in K} y^\top x,$$

and $y \in K_*$ follows from the second inequality. But also the first of these is equivalent to

$$c^\top x \leq P^* \Rightarrow (A^\top y)^\top x \leq y^\top b.$$

This can only happen if $A^\top y = \alpha c$ and $y^\top b \geq \alpha P^*$ for some $\alpha \geq 0$. For the final bit of casework: If $\alpha > 0$, then take $y^* = \frac{y}{\alpha} \in K_*$ for which $A^\top y^* = c$ and $b^\top y^* \geq P^*$. We are done.

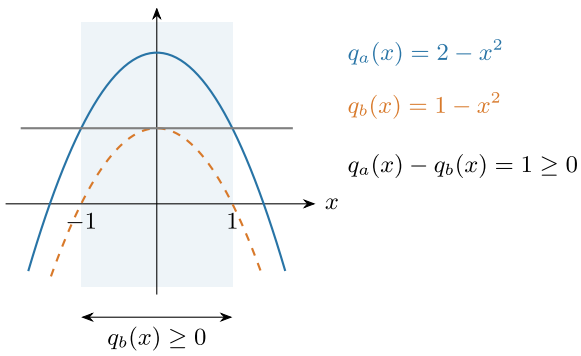
If $\alpha = 0$, $A^\top y = 0$ and $b^\top y \geq 0$. Note that there is an x such that $Ax \underset{K}{>} b$. Because $y \in K_*$ is not zero, we have $y^\top (Ax - b) > 0$, but that implies $y^\top Ax = 0 > b^\top y$, which is a contradiction. □

18. The S-Lemma

Consider a non-convex QCQP $\min_x q_a(x) = x^\top Q_a x + q_a^\top x + c_a$ subject to the quadratic constraint $q_b(x) = x^\top Q_b x + q_b^\top x + c_b \geq 0$. Here Q_a, Q_b are arbitrary; it need not be $Q_a \succeq 0$, $Q_b \succeq 0$.

Such programs are efficiently solvable, despite apparent non-convexity.

Theorem 65 Let there exist x such $q_b(x) > 0$ strictly. Then $\forall x, q_b(x) \geq 0 \Rightarrow q_a(x) \geq 0$ implies $\exists \lambda \geq 0$ such that for all x, $q_a(x) \geq \lambda q_b(x)$.



$\Leftarrow$ above is evident. Such λ "certifies" the former clause. This has the same spirit as Farkas' lemma or, theorem of alternatives, as you might have seen before:

$$
\begin{aligned}
&\min_x q_a(x) \;\text{ subject to }\; q_b(x) \geq 0 \\
\equiv &\max_\gamma \gamma \;\text{ subject to }\; \forall x, q_b(x) \geq 0 \Rightarrow q_a(x) - \gamma \geq 0 \\
\equiv &\max_{\gamma,\lambda\geq 0} \gamma \;\text{ subject to }\; \forall x, q_a(x) - \gamma \geq \lambda q_b(x) && \text{(using S-lemma)} \\
\equiv &\max_{\gamma,\lambda\geq 0} \gamma \;\text{ subject to }\; \forall x, \begin{pmatrix} x \\ 1 \end{pmatrix}^\top \begin{pmatrix} Q_a - \lambda Q_b & \frac{q_a - \lambda q_b}{2} \\ \frac{(q_a-\lambda q_b)^\top}{2} & c_a - \gamma - \lambda c_b \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} \geq 0 \\
\equiv &\max_{\gamma,\lambda\geq 0} \gamma \;\text{ subject to }\; \begin{pmatrix} Q_a - \lambda Q_b & \frac{q_a - \lambda q_b}{2} \\ \frac{(q_a-\lambda q_b)^\top}{2} & c_a - \gamma - \lambda c_b \end{pmatrix} \succeq 0
\end{aligned}
$$

This needs a brief justification. In terms of feasible sets, $\supseteq$ is clear. You might reasonably suspect we are asking for more than what is necessary, e.g., $(x,0)^\top M(x,0) \geq 0$ also, in the latter formulation. To complete the argument, say there exists y, $y^\top My < 0$. Then,

1. last coordinate of $y \neq 0$, then by dividing by scaling there exists x such that $(x,1)^\top M(x,1) < 0$.
2. last coordinate of $y = 0$, since $y^\top My$ is continuous, there exists z with non-zero last coordinate with $z^\top Mz < 0$. We are back to the case above.

18.1. The Homogenous Case

We will work towards proving the S-lemma. Let's start with something simpler.

Lemma 66 For symmetric A, B, $\operatorname{tr}(A) \geq 0 > \operatorname{tr}(B) \Rightarrow \exists x, x^\top Ax \geq 0$ and $x^\top Bx < 0$.

Proof. Two useful observations: For any $x \in \{-1,1\}^n$, $x^\top Dx = \operatorname{tr}(D)$ for any diagonal D, since $x_i^2 = 1$. Secondly, for any matrix M, $\mathbb{E}_{x\sim \text{ Unif}(\{-1,1\}^n)}[x^\top Mx] = \operatorname{tr}(M)$.

Now choose $\xi \sim \text{Unif}(\{-1,1\}^n)$, $x = U\xi$, where $B = U\Lambda U^\top$ diagonal.

$$x^\top Bx = \xi^\top U^\top U\Lambda U^\top U\xi = \operatorname{tr}(\Lambda) = \operatorname{tr}(\Lambda U^\top U) = \operatorname{tr}(B) < 0$$

$$\mathbb{E}[x^\top Ax] = \mathbb{E}\xi^\top (U^\top AU)\xi = \operatorname{tr}(U^\top AU) = \operatorname{tr}(AUU^\top) = \operatorname{tr}(A) \geq 0.$$

Therefore, there exists x such $x^\top Ax \geq 0$ and $x^\top Bx < 0$. □

The following is a simplified version of the S-lemma without the linear or intercept terms.

Theorem 67 Say $\exists \overline{x}, \overline{x}^\top A\overline{x} > 0$. Then $\forall x, x^\top Ax \geq 0 \Rightarrow x^\top Bx \geq 0$ implies $\exists \lambda \geq 0, B \succeq \lambda A$.

Proof. Consider $\text{OPT}' = \min_{x:x^\top Ax\geq 0, x^\top x = n} x^\top Bx$, which we can relax to an SDP as

$$
\begin{aligned}
\text{OPT}' &\equiv \min_X \operatorname{tr}(BX) \;\text{ subject to }\; \operatorname{tr}(AX) \geq 0, \operatorname{tr}(X) = n, X \succeq 0, X \;\text{ is rank 1.} \\
&\geq \min_X \operatorname{tr}(BX) \;\text{ subject to }\; \operatorname{tr}(AX) \geq 0, \operatorname{tr}(X) = n, X \succeq 0 && \text{(SDP)} \\
&\equiv \max_{u\in\mathbb{R},\lambda\geq 0} nu \;\text{ subject to }\; B \succeq uI + \lambda A. && \text{(dual)}
\end{aligned}
$$

Exercise. Show that the SDP dual is indeed the dual of the SDP in the last step.

Let's say we guarantee that SDP ≥ 0 and that the dual is solvable and strong duality holds. Then, we are done as then there exist $u, \lambda \geq 0$ such $B \succeq uI + \lambda A$ and hence $B \succeq \lambda A$.

For small enough $\alpha > 0$, we can guarantee that

$$\overline{X} = \frac{\overline{xx}^\top + \alpha I}{\mathrm{tr}(\overline{xx}^\top + \alpha I)} n,$$

is strictly feasible for the SDP. And for all $\lambda > 0$, there exists some negative enough u such that $B \succ uI + \lambda A$. Thus the SDP dual is also strictly feasible and strong duality holds. (Why?)

Now to argue SDP ≥ 0, compactness ensures that it is solvable; let X^* be the solution. There exists L such that $X^* = LL^\top$. Consider that

$$\mathrm{tr}(AX^*) = \mathrm{tr}(L^\top AL) \geq 0, \text{ and } \mathrm{tr}(BX^*) = \mathrm{tr}(L^\top BL) = nu^*.$$

If $u^* \geq 0$, we are done. If $u^* < 0$, then by our sign lemma, there exists x such $(Lx)^\top A(Lx) \geq 0$ and $(Lx)^\top B(Lx) < 0$. But this contradicts $\forall x, x^\top Ax \geq 0 \Rightarrow x^\top Bx \geq 0$. □

Exercise. In words, describe why adding $x^\top x = n$ as a constraint was essential for this argument.

Can you think of an alternative proof that works without adding $x^\top x = n$?

18.2. Finishing up the General Case

Proof of the original theorem. The idea is to use homogeneous polynomials.

$$\overline{q}_a(x,t) = x^\top Q_a x + q_a^\top xt + c_a t^2.$$

$$\overline{q}_b(x,t) = x^\top Q_b x + q_b^\top xt + c_b t^2.$$

Given there exists x with $q_b(x) > 0$, we have $\overline{q}_b(x,1) > 0$.

We'll prove $\forall (x,t), \overline{q}_b(x,t) \geq 0 \Rightarrow \overline{q}_a(x,t) \geq 0$. If so, using our simplified S-lemma, we have that there exists $\lambda \geq 0$, $\overline{q}_a(x,t) \geq \lambda \overline{q}_b(x,t)$ for all (x,t). Set $t = 1$ to finish the proof.

Let's say our sub-claim is untrue. Then, $\exists (x,t)$ such that $\overline{q}_b(x,t) \geq 0$ but $\overline{q}_a(x,t) < 0$. Let us consider the three possibilities:

- $t \neq 0$. Then substituting $\left(\frac{x}{t}, 1\right)$ results in contradiction.
- $t = 0$, $\overline{q}_b(x,t) > 0$, by continuity there exists (y,s) with $s \neq 0$ and $\overline{q}_b(y,s) \geq 0$. Case (1) applies.
- $t = 0$, $\overline{q}_b(x,t) = 0$. It must then be that $x^\top Q_b x = 0$ and $x^\top Q_a x < 0$. Increase t slightly in the direction of $q_b^\top x$ (zero means anything works), so that $\overline{q}_b(x,t) \geq 0$, while retaining $\overline{q}_a(x,t) < 0$. This only fails if $q_b^\top x = 0$ and $c_b < 0$, in which case, after slightly increasing t, scaling up x enough can guarantees $\overline{q}_b(x,t) \geq 0$ and $\overline{q}_a(x,t) < 0$ with a non-zero t. Case (1) applies again.

□

47-851: Modern Convex Optimization Lecture 4

Volumetric Algorithms

Lecturer: Karan Singh

47-851: Modern Convex Optimization Lecture 5

First-Order Methods

Lecturer: Karan Singh

47-851: Modern Convex Optimization Lecture 6

Assignment 1

Lecturer: Karan Singh

Q1. Inner Products and Dual Norms (10 points)

Part A. Prove that a norm on a real vector space V is induced by an inner product if and only if it satisfies the parallelogram identity

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \quad \forall x, y \in V.$$

Part B. For a norm $\|\cdot\|$ on $\mathbb{R}^n$, define its dual norm as

$$\|x\|_* = \max_{\|y\| \le 1} y^\top x.$$

1. Calculate the dual norms of $\|\cdot\|_2$ and $\|\cdot\|_\infty$, where $\|x\|_2 = \sqrt{x^\top x}$ and $\|x\|_\infty = \max_i |x_i|$.
2. Show that

$$\partial \|\cdot\|(x) = \{y : \|y\|_* \le 1,\ y^\top x = \|x\|\}.$$

Q2. Local Descent and Checking PSD (10 points)

Part A. Find a function $f : \mathbb{R}^n \to \mathbb{R}$ and a point $x \in \mathbb{R}^n$ such that there is no (non-zero) descent direction at x, and yet x is not a local minimizer.

Part B. Given $x \in \mathbb{R}^n$, form a $(n+1) \times (n+1)$-dimensional matrix

$$X = \begin{bmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ x_1 & x_1 & 0 & \cdots & 0 \\ x_2 & 0 & x_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & 0 & 0 & \cdots & x_n \end{bmatrix}.$$

Show that $X \succeq 0$ if and only if $x_i \ge 0, \forall i \in [n]$ and $\sum_{i=1}^n x_i \le 1$.

Q3. Quasiconvexity and Back (10 points)

Part A. Prove that $f(x) = \log\left(\sum_{i=1}^n a_i e^{x_i}\right)$ where $a_i > 0$ is convex.

Part B. Call a function *kinda-convex* if all its sublevel sets are convex.

1. Give a function that is kinda-convex but not convex.
2. Prove that, for any convex $f : \mathbb{R} \to \mathbb{R}$ and nondecreasing $g : \mathbb{R} \to \mathbb{R}$, $f(g(x))$ is kinda-convex.

3. Can such functions be made convex by stretching the x-axis? More precisely, given a kinda-convex $h : \mathbb{R} \to \mathbb{R}$, does there exist a differentiable, invertible $g : \mathbb{R} \to \mathbb{R}$ such that $h(g(x))$ is convex? Prove your claim.

Challenge (Zero Credits): Try to generalize (A) to $f(X_1, ..., X_n) = \log \operatorname{tr}\left(\sum_{i=1}^{n} a_i e^{X_i}\right)$. Matrix exponential simply exponentiates the eigenvalues in the spectral decomposition.

47-851: Modern Convex Optimization Lecture 7

Assignment 2

Lecturer: Karan Singh

Q4. Eigenvalues and SDPs (15 points)

Part A. Let $a_i \sim \mathcal{N}(\mu_i, \Sigma_i)$ be multivariate normal random vectors. Recast the following problem as an SDP assuming $1/2 \leq \eta < 1$:

$$\max_x c^T x \quad \text{subject to} \quad \Pr(a_i^T x \leq b_i) \geq \eta \quad (i = 1, ..., m).$$

Part B. Recast the following problem as an SDP with strict inequality constraints allowed:

$$\min_X \frac{\lambda_{\max}(X)}{\lambda_{\min}(X)} \quad \text{subject to} \quad \langle A_i, X \rangle = b_i \quad (i = 1, ..., m), \quad X \succ 0.$$

Part C. Recast the following problem as an SDP:

$$\min_{X \in \mathbb{S}^{n \times n}} \sum_{j=1}^{k} \lambda_j(X) \quad \text{subject to} \quad \langle A_i, X \rangle = b_i \quad (i = 1, ..., m),$$

where $\lambda_1(X) \geq ... \geq \lambda_n(X)$ are the eigenvalues.

Hint: Recast $\max_{U \in \mathbb{R}^{n \times k}, U^\top U = I} \operatorname{tr}(U^\top X U) = \sum_{i=1}^{k} \lambda_i(X)$ as an SDP and consider the dual.

Q5. Lovász's bound on Shannon capacity (15 points)

Let $\alpha(G)$ be the maximum size of an independent set. Recall

$$\Theta(G) = \lim_{k \to \infty} \sqrt[k]{\alpha(G^k)} = \sup_{k \geq 1} \sqrt[k]{\alpha(G^k)}.$$

Part A. Let J be the all-ones matrix and define

$$\nu(G) = \max_X \big\{ \operatorname{tr}(JX) : \operatorname{tr}(X) = 1,\ X \succeq 0,\ X_{ij} = 0 \text{ if } \{i,j\} \in E \big\}.$$

Show that $\nu(G) \geq \alpha(G)$.

Part B. Show that $\alpha(G_A \boxtimes G_B) \geq \alpha(G_A)\alpha(G_B)$. The strong product has vertex set $V_A \times V_B$. Distinct vertices $(i,k), (j,\ell)$ are adjacent exactly when

$$(\{i,j\} \in E_A \text{ or } i = j) \quad \text{and} \quad (\{k,\ell\} \in E_B \text{ or } k = \ell).$$

Part C. Show that $X, Y \succeq 0$ implies $X \otimes Y \succeq 0$. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{p \times q}$, the Kronecker product $A \otimes B$ is defined as

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mn}B \end{bmatrix}.$$

Part D. Show that $\nu(G_A \boxtimes G_B) \leq \nu(G_A)\nu(G_B)$.

Hint: Use the dual SDP and try to *Kronecker* two feasible solutions using part C.

Part E. Show that $\nu(G) \geq \Theta(G)$ and use this to show $\nu(C_5) = \Theta(C_5) = \sqrt{5}$.

Hint: Numerically solving the SDP to see if it is close enough to $\sqrt{5}$ is an acceptable proof.

47-851: Modern Convex Optimization Lecture 8

Assignment 3

Lecturer: Karan Singh

Q6. Grads and Dists (10 points)

Part A: Consider a function $f : \mathbb{R}^d \to \mathbb{R}$ that is μ-strongly convex and L-smooth. Let $x^* \in \arg\min_x f(x)$. Prove for all $x \in \mathbb{R}^d$ that

$$\frac{1}{2L}\|\nabla f(x)\|^2 \leq f(x) - f(x^*) \leq \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Part B. Prove that f is μ-strongly convex if and only if for all $x, y \in \mathbb{R}^d$, we have

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \mu \|x - y\|^2.$$

Q7. Just Enough Convexity (10 points)

Consider a L-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ that satisfies for all x that

$$f(x) - \min_{x^*} f(x^*) \leq \frac{1}{2\mu}\|\nabla f(x)\|^2.$$

Part A: Prove that such a function may not even be convex (much less strongly convex).

Part B: Nevertheless, prove that gradient descent on such a function, with an appropriate step size, produces iterates so that

$$f(x_t) - \min_{x^*} f(x^*) \leq O\Big(e^{-\mu \frac{t}{L}}\Big).$$

Q8. Small Grad Bounds (10 points)

Part A: Fix a convex set $K \subset \mathbb{R}^d$ of diameter D. Recall that online gradient guarantees the following regret bound against a sequence of adversarially chosen convex G-Lipschitz functions f_t.

$$\text{Regret} = \sum_{t=1}^{T} f_t(x_t) - \min_{x^* \in K} \sum_{t=1}^{T} f_t(x^*) \leq O\Big(GD\sqrt{T}\Big),$$

With suitable dynamically chosen step sizes, show that one can guarantee that

$$\text{Regret} \leq O\left(D\sqrt{\sum_{t=1}^{T} \|\nabla f_t(x_t)\|^2}\right).$$

Hint: Set $\eta_t \propto \frac{1}{\sqrt{\sum_{s=1}^{t} \|\nabla f_s(x_s)\|^2}}$.

Part B: Assume that all loss functions $f_t : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ are in addition non-negative, well-defined and L-smooth over all of $\mathbb{R}^d$. Our online algorithm must still play points in K and the regret is still defined with respect to the best point in K. Prove that the same algorithm guarantees

$$\text{Regret} \leq O\left(\sqrt{LD^2\left(\min_{x^* \in K} \sum_{t=1}^{T} f_t(x^*)\right)} + LD^2\right).$$

This means whenever there is a point with zero net loss, we get constant regret, independent of T.

Hint: Use part A from the first question.